

Complementary Therapies in Medicine

EFFECTS OF AUTOMATED MASSAGE CHAIR THERAPY ON MENTAL HEALTH AND PHYSICAL HEALTH: A COMPREHENSIVE STUDY

--Manuscript Draft--

Manuscript Number:	CTIM-D-25-01985
Article Type:	VSI: Physical activity and health (Original Research)
Section/Category:	Other
Keywords:	hypertension; low back pain; massage chair; massage fatigue; stress
Corresponding Author:	Lai Khin Wee Universiti Malaya Department of Biomedical Engineering MALAYSIA
First Author:	XIN LEE
Order of Authors:	XIN LEE JULIANA USMAN NORHAMIZAN HAMZAH PAULINE SHAN QING YEOH XIANG WU JOON HUANG CHUAH Lai Khin Wee
Abstract:	This 3-part study evaluated the effects of automated massage chair therapy on mental and physical health. In Part 1, 20 moderately stressed students participated in a randomized crossover study, receiving either a 20-minute automated massage chair therapy session followed by a 20-minute control session, or vice versa, with a 48-hour washout period. Blood pressure (BP), heart rate (HR), electroencephalogram (EEG), State-Trait Anxiety Inventory (STAI), and Visual Analog Scale (VAS) were measured to assess stress, anxiety, and mental fatigue. The results showed that automated massage chair therapy significantly reduced diastolic BP, HR, stress, and anxiety compared to the control intervention. In contrast, mental fatigue was reduced following both interventions, with no significant difference between them. In Part 2, 20 hypertensive hospital staff in a one-group pretest-posttest study received three 20-minute automated massage chair therapy sessions on alternate days. BP, HR, and skin blood flow (SBF) were measured, and the results showed that BP and HR decreased significantly, while SBF did not change significantly. In Part 3, 20 hospital staff with chronic low back pain participated in a one-group pretest-posttest study, receiving three 20-minute automated massage chair therapy sessions on alternate days. Electromyogram (EMG) and VAS were measured, and the results showed an improvement in low back function (EMG) and a significant reduction in low back pain (VAS). These results highlight the potential of automated massage chair therapy as an effective method for reducing stress, lowering blood pressure, and alleviating low back pain.
Opposed Reviewers:	

Lai Khin Wee
Department of Biomedical Engineering,
Universiti Malaya, 50603 Kuala Lumpur, Malaysia

Dr. Abid Hasan Khan
Guest Editor
Physical activity for physical and/ or mental health
Complementary Therapies in Medicine

24 September 2025

Dear Dr. Abid Hasan Khan,

On behalf of my co-authors, I am pleased to submit our manuscript entitled “Effects of Automated Massage Chair Therapy on Mental Health and Physical Health: A Comprehensive Study” for consideration in the special issue: Physical activity for physical and/ or mental health in Complementary Therapies in Medicine.

In today’s fast-paced world, physical and mental health are constantly under discussion, and while regular exercise is well-established as beneficial, many individuals face barriers such as limited motivation or reduced physical capability. To address these challenges, complementary and technology-assisted intervention may provide potential solutions, but further evidence is needed to support their effectiveness. Therefore, this comprehensive study evaluated the effects of automated massage chair therapy on both mental and physical health as an emerging modality for improving well-being. Our findings are significant because the results indicated that automated massage chair therapy can reduce stress, anxiety, mental fatigue, blood pressure, and pain. We believe that these improvements will translate to higher motivation, more energy, and enhance overall well-being, thereby allowing more individuals to pursue and maintain an active lifestyle for long-term health benefits.

In conclusion, automated massage chair therapy represents a practical, accessible modality that can support both mental and physical well-being, particularly for individuals who may find traditional exercise challenging. Looking forward, technological advancements such as AI-driven personalized massage programs could further enhance efficacy and engagement. Widespread implementation of such devices in public spaces, including universities and hospitals, has the potential to improve health equity by providing accessible support for both mental and physical health across diverse populations.

We believe our work offers valuable insights into innovative, technology-assisted strategies for promoting well-being and would be of interest to your readership. We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.

Thank you for your valuable time. We look forward to the opportunity to contribute to this Special Issue.

Sincerely,
Lai Khin Wee
Department of Biomedical Engineering,
Universiti Malaya, 50603 Kuala Lumpur, Malaysia
lai.khinwee@um.edu.my

- Automated massage chair therapy reduced stress, anxiety, and mental fatigue for students.
- Automated massage chair therapy lowered blood pressure for hypertensive individuals.
- Automated massage chair therapy reduced low back pain.
- Automated massage chair therapy is well tolerated by all participants.
- All the participants complete the study without developing any side effects.

Effects of Automated Massage Chair Therapy on Mental Health and Physical Health: A Comprehensive Study

Xin Lee ¹, Juliana Usman ¹, Norhamizan Hamzah ², Pauline Shan Qing Yeoh ¹, Xiang Wu ^{4,*},
Joon Huang Chuah ³, Khin Wee Lai ^{1,*}

¹ Department of Biomedical Engineering, Universiti Malaya, 50603 Kuala Lumpur, Malaysia

² Department of Rehabilitation Medicine, Universiti Malaya, 50603 Kuala Lumpur, Malaysia

³ Department of Electrical Engineering, Universiti Malaya, 50603 Kuala Lumpur, Malaysia

⁴ Institute of Medical Information Security, Xuzhou Medical University, Xuzhou 221004, China

* Corresponding author at: Department of Biomedical Engineering, Universiti Malaya, 50603 Kuala Lumpur, Malaysia

E-mail address: lai.khinwee@um.edu.my

Institute of Medical Information Security, Xuzhou Medical University, Xuzhou 221004, China

E-mail address: wuxiang@xzhmu.edu.cn

This 3-part study evaluated the effects of automated massage chair therapy on mental and physical health. In Part 1, 20 moderately stressed students participated in a randomized crossover study, receiving either a 20-minute automated massage chair therapy session followed by a 20-minute control session, or vice versa, with a 48-hour washout period. Blood pressure (BP), heart rate (HR), electroencephalogram (EEG), State-Trait Anxiety Inventory (STAI), and Visual Analog Scale (VAS) were measured to assess stress, anxiety, and mental fatigue. The results showed that automated massage chair therapy significantly reduced diastolic BP, HR, stress, and anxiety compared with the control intervention. In contrast, mental fatigue was reduced following both interventions, with no significant difference between them. In Part 2, 20 hypertensive hospital staff in a one-group pretest-posttest study received three 20-minute automated massage chair therapy sessions on alternate days. BP, HR, and skin blood flow (SBF) were measured, and the results showed that BP and HR decreased significantly, while SBF did not change significantly. In Part 3, 20 hospital staff with chronic low back pain participated in a one-group pretest-posttest study, receiving three 20-minute automated massage chair therapy sessions on alternate days. Electromyogram (EMG) and VAS were measured, and the results showed an improvement in low back function (EMG) and a significant reduction in low back pain (VAS). These results highlight the potential of automated massage chair therapy as an effective method for reducing stress, lowering blood pressure, and alleviating low back pain.

Keywords: hypertension; low back pain; massage chair; mental fatigue; stress.

1. Introduction

These days, students must compete intensively to secure admission to top-ranked universities. This behavior highlights the concept of ‘academic involution’, as students engage in excessive and often irrational over-competition for limited, high-quality resources [1]. In this environment, students experience constant pressure from peers, family, and society to excel academically; in fact, research shows a positive correlation between academic involution and student stress levels [2]. In China, a study involving 11,954 students sampled from 50 Chinese universities showed that more than one-third (36.9%) of the participants experienced severe mental stress [3]. In reality, high levels of stress are not limited to academic settings but are also prevalent in healthcare environments; for instance, a cross-sectional study in Japan showed that healthcare workers experience higher stress levels than the national average [4].

Stress refers to a state of tension, anxiety, or distress that arises in response to challenging or threatening situations. It is generally categorized as acute or chronic: acute stress is short-term and triggered by specific events, while chronic stress develops from long-term pressures and ongoing challenges. The physiological response to such threats, known as the “fight or flight” response, involves the activation of the hypothalamic-pituitary-adrenal (HPA) axis, which triggers the release of cortisol that raises blood sugar and increases alertness to help the body respond effectively [5]. In school settings, short-term stress can temporarily enhance concentration and cognitive function, allowing students to process information more efficiently [6, 7]. However, an extended period of heightened stress may overwhelm the individual, increasing the risk of developing anxiety and mental fatigue [8, 9]. From a neuroendocrine standpoint, chronic stress causes persistent activation of the hypothalamic-pituitary-adrenal (HPA) axis, resulting in sustained cortisol secretion and hormonal imbalance that disrupts the gamma-aminobutyric acid (GABA) system [10]. Impairment of the GABA system diminishes its inhibitory and calming effects, leading to over-excitation in the brain and a reduced ability to control emotions, which can manifest as anxiety symptoms such as excessive worry, restlessness, and impaired concentration. In addition to anxiety, chronic academic stress can contribute to the development of mental fatigue, which is characterized by decreased concentration, impaired working memory, and reduced cognitive flexibility [11]. Almarzouki [12] shows that mental fatigue impairs working memory, thereby reducing the brain’s capacity to hold and manipulate information, which in turn contributes to poorer academic performance. Furthermore, constant pressure and sustained mental effort can lead to burnout, a more severe and persistent state of emotional, mental, and physical exhaustion [13]. Therefore, adequate rest is vital to prevent the accumulation of mental fatigue. However, research has shown that short breaks during a seven-hour workday are insufficient to prevent mental fatigue [14], suggesting that passive rest alone may not be effective and highlighting the potential benefit of incorporating activities such as brief exercise during breaks.

The impact of stress extends beyond mental health and has a profound effect on physical health. Chronic stress can significantly impair physical health by disrupting multiple bodily systems, including cardiovascular, immune, digestive, endocrine, and musculoskeletal systems. The cardiovascular system is particularly vulnerable to stress exposure, and cardiovascular diseases remain the leading cause of death worldwide. In 2017, an estimated 17.8 million people died from cardiovascular disease, and high blood pressure was the leading cause of death in cardiovascular disease [15]. In stressful situations, the adrenal glands release adrenaline and noradrenaline, which act swiftly to increase the heart rate, enhance blood flow, and temporarily elevate blood pressure. However,

continuous exposure to stress can lead to hypertension, increasing the risk of cardiovascular issues such as heart attack and stroke [16]. In a Black community cohort, higher perceived stress over time is associated with a significantly higher risk of developing hypertension [17]. Aside from cardiovascular disease, chronic stress can also manifest in a range of other physical symptoms. Haight, Peddie [18] reported that sustained stress in middle-aged adults is associated with increased severity and frequency of physical symptoms, including fever, allergies, headaches, and muscle soreness. Stress also impairs the musculoskeletal system by causing muscle tension, aches, and pain in the neck, shoulders, or lower back. A cross-sectional survey involving medical students found a moderate positive correlation between stress and the incidence of musculoskeletal disorders [19], while Gilchrist and Pokorná [20] discovered that 85% of registered nurses experienced a high incidence of low back pain over the past 12 months. One key explanation is that the sudden onset of stress triggers the “fight or flight” response, causing the body to tighten its muscles in preparation for immediate physical action [21]. Over time, sustained activation of these muscles can lead to spasm, stiffness, discomfort, and pain. In addition, individuals under stress often adapt to poor posture, such as slouching and hunching. This places additional strain on the spine, contributing to musculoskeletal issues such as low back pain. In addition, chronic stress is also known to alter how the brain processes pain signals, often amplifying the perception of pain. This phenomenon is known as central sensitization, and it causes several chronic pain conditions such as fibromyalgia, chronic low back pain, and temporomandibular joint (TMJ) disorder [22-24].

Managing stress is important because chronic stress can harm both mental and physical health. Antidepressants are commonly prescribed to address mental health issues such as anxiety and depression, but medication non-adherence remains a prevalent issue. In Castile and Leon, Spain, nearly one-fifth of the population on antidepressants did not adhere to their medication regimen, primarily due to medication side effects [25]. Likewise, hypertension management relies heavily on antihypertensive medications, but many patients struggle with adherence due to side effects such as dizziness, fatigue, and sexual dysfunction, in addition to the asymptomatic nature of the condition. Low back pain, on the other hand, is often managed with non-steroidal anti-inflammatory drugs or painkillers to alleviate pain and reduce inflammation; however, these medications may risk damaging the cardiovascular, renal, and gastrointestinal systems. The American College of Physicians recommends non-drug treatments as the first option for acute or subacute low back pain, due to the risks associated with medications. This has led to an increased interest in complementary and alternative medicine (CAM) due to its minimal side effects. Examples of CAM therapies include music therapy, hypnotherapy, cognitive behavioral therapy, animal-assisted therapy, aromatherapy, massage therapy, meditation, and yoga, all of which have demonstrated effectiveness in reducing stress, anxiety, and depression [26-33]. Among these, massage therapy, which involves the manipulation of soft tissues such as muscles, tendons, ligaments, and skin, has been shown to improve blood flow, facilitate healing, and induce relaxation. Massage therapy typically involves the hands-on manipulation of soft tissue, using light to medium pressure, as seen in Swedish massage, or deeper pressure, as found in deep tissue massage or sport massage. Different pressure levels target specific tissue layers, providing therapeutic benefits based on individual needs. Massage therapy works by stimulating nerve endings to reduce pain and promote relaxation. Earlier studies suggest it increases parasympathetic activity while decreasing sympathetic nervous system activation [34]. Nelson [35] discovered that an increase in parasympathetic activity can lower blood pressure and increase heart rate variability. Furthermore, massage can enhance athletic performance by increasing blood circulation [36], as confirmed by laser Doppler flowmetry showing elevated blood flow following massage [37].

The mechanism underlying massage therapy can be explained primarily through three pathways: mechanical effects, biochemical processes, and the gate-control theory of pain. The mechanical effects occur as friction-induced heat causes blood vessels to dilate, improving blood circulation. This enhanced circulation improves the delivery of oxygen and nutrients to tissues, promotes tissue regeneration, and facilitates the removal of metabolic waste such as lactic acid. A study by Zhang, Xu [38] revealed that individuals with chronic wounds who received Chinese massage twice weekly for 12 weeks experienced a 30% increase in lesion healing rates, demonstrating a positive impact of massage on tissue healing and regeneration. In addition to its mechanical effects, massage therapy also influences body chemistry. The application of mechanical pressure to the skin stimulates the mechanoreceptors, which send signals to the vagus nerve, activating the parasympathetic nervous system. The activation of the parasympathetic nervous system helps lower stress hormones such as cortisol while simultaneously increasing the production of feel-good hormones, including dopamine and serotonin [39]. In fact, an increase in serotonin levels is associated with less pain and fewer sleep disturbances among adults with

migraine headache [40]. These biochemical changes contribute to reduced stress and anxiety, improved mood, and pain reduction. The third mechanism of massage therapy involves the gate-control theory of pain, proposed by Melzack and Wall [41]. The gate-control theory of pain is a well-known neurological model that explains how non-painful stimuli can inhibit the perception of painful stimuli. This theory suggests that non-painful stimuli, such as mechanical pressure from massage, can 'close the gate' to painful stimuli, preventing them from reaching the brain. During massage therapy, large-diameter nerve fibers are activated to transmit touch and pressure signals, while the transmission of pain signals via small-diameter nerve fibers is inhibited. This selective inhibition of painful signals makes massage particularly effective for managing pain in different body parts.

Massage therapy has a rich history spanning thousands of years, with origins traced back to ancient civilizations such as Egypt, Greece, and China. Over time, it has evolved into different forms, including Swedish massage, Thai massage, deep tissue massage, and aromatherapy massage. Building on these traditional practices, modern technology has introduced automated massage chairs, which have become increasingly popular due to their convenience, consistency, and cost-effectiveness [42]. Therefore, this study aims to find out the effects of automated massage chair therapy on mental and physical health based on electroencephalography (EEG), electromyography (EMG), skin blood flow, blood pressure, heart rate, and self-reported psychological measures such as State-Trait Anxiety Inventory (STAI) and Visual Analog Scale (VAS). The study is divided into three parts: (1) effects of automated massage chair therapy on stress, anxiety, and mental fatigue; (2) effects of automated massage chair therapy on blood pressure, heart rate, and skin blood flow; (3) effects of automated massage chair therapy on low back function and pain.

2. Part 1: Effects of Automated Massage Chair Therapy on Stress, Anxiety, and Mental Fatigue

2.1. Study Design

This randomized, 2x2 crossover pilot trial compared the effects of automated massage chair therapy and control treatment among moderately stressed students from Universiti Malaya. The study took place at the Neuroimaging Laboratory, Faculty of Engineering, Universiti Malaya. Participants were assigned to massage treatment followed by control treatment, or control treatment followed by massage treatment, with a 48-hour washout period.

2.2. Participants and Recruitment

Participants were recruited using convenience sampling through an invitation email sent via the Universiti Malaya student mailing list. The mailing list included students from diverse academic disciplines, ethnic backgrounds, and nationalities. This recruitment strategy increased the likelihood of obtaining a heterogeneous sample and increased the generalizability of the findings beyond the university context. The invitation email included an overview of the study and a Perceived Stress Scale (PSS) questionnaire. PSS is a validated instrument for evaluating subjective stress levels. The participants were fully informed about the study procedures, as well as the potential benefits and risks of the intervention. Participation in the study was completely voluntary, and participants were free to withdraw at any time without consequence. All participants provided informed written consent before data collection. The eligibility criteria were as follows: 1) moderate to high stress level (PSS scores > 14), 2) age between 18-35 years old, 3) able to participate for the entire study duration, 4) not pregnant, 5) not receiving concurrent massage therapy, 6) no active psychiatric condition or neurological condition. The study was reviewed and approved by the University of Malaya Research Ethics Committee (Reference No. UM.TNC2/UMREC_3259)

2.3. Study Procedure and Intervention

The overall experimental procedure for the study is outlined in Figure 2.1. Every participant completed two separate sessions: one automated massage chair therapy session and one control treatment session. The order of the two sessions was randomized to minimize the potential sequence effects. The overall experimental procedure can be divided into: a) 10-minute acclimatization period, b) baseline assessment, c) mental fatigue protocol, d) 20-minute treatment, e) post-intervention assessment.

Before every session, participants were instructed to sit on the massage chair for 10 minutes to acclimatize to the environment. The baseline assessment included measurements such as STAI, VAS, blood pressure, heart rate, and EEG. The mental fatigue protocol included a 10-minute cognitively demanding test and EEG recording. Afterward, participants received either automated massage chair therapy or control treatment for 20 minutes. The session concluded with a post-intervention assessment using the same measurements as the baseline assessment. The EEG activity was measured at three time points: immediately before the cognitive test, immediately after the cognitive test, and immediately after the treatment. The blood pressure was measured immediately before EEG measurement during the baseline assessment and three minutes after the treatment during the post-intervention assessment. Participants were instructed to minimize movement and refrain from falling asleep during the experiment. Moreover, they were asked to refrain from consuming caffeine and alcohol and to ensure they had adequate sleep the night before the experiment. The demographic characteristics of the study participants were collected using a questionnaire on their first visit.

2.4. Equipment and Outcome Measures

2.4.1. Ogawa Master Drive AI 2.0 (OG-7598X) Massage Chair

The automated massage chair therapy was delivered using Ogawa Master Drive AI 2.0 (OG7598X, Ogawa, Malaysia) equipped with 64 air compression chambers. The pre-programmed massage mode (Master's Choice massage program) was used to massage the whole body for 20 minutes during the massage treatment. During the control treatment, the participant remained seated on the massage chair for 20 minutes.

2.4.2. Electroencephalogram

The participants were seated on the automated massage chair during the EEG measurement. EEG data were acquired using a 20-channel Medelec Profile system (Medelec, Oxford Instruments, UK). Two electrodes (Fp1 and Fp2) were placed on the frontopolar region according to the international 10-20 system. The ground electrode and reference electrode were attached to the left and right mastoids respectively. Electrode impedances were maintained below $1k\Omega$ by using the electro-conductive Ag/AgCl electrodes and SignaGel (Parker Laboratories, US). The skin was prepared in advance with Nuprep (Weaver and Company, United States) before electrode placement to reduce impedance. EEG activity was recorded for 3 minutes while the participants remained completely still on the massage chair with their eyes closed as shown in Figure 2.2. The data were acquired at a sampling rate of 512 Hz and filtered using a 0.5-70 Hz band-pass filter with a 50 Hz notch filter.

MATLAB R2024b was used for data preprocessing, including filtering, segmentation, and artifact removal. The recordings were further processed offline using a 0.5 Hz high-pass and a 40 Hz low-pass filter. The recordings were then segmented into 2-second epochs, and any epochs containing artifacts were manually rejected. To improve signal quality, Independent Component Analysis (ICA) was applied to remove artifacts caused by muscle movements, eye blinks, or eye movements. Power spectrum analysis was performed on the artifact-free data using Fast Fourier Transform (FFT) algorithms. The mean power values for the delta, theta, alpha, and beta frequency bands were calculated by averaging the values within each band. The mental fatigue index was calculated using the formula $(\text{theta} + \text{alpha}) / \text{beta}$.

2.4.3. Blood Pressure and Heart Rate

Blood pressure and heart rate were measured using an Omron automatic blood pressure monitor HEM-7156T (Omron, Japan). The blood pressure cuff was placed on the left upper arm 2.5 cm above the antecubital fossa. Three readings were taken at 30-second intervals, and the average of the three measurements was calculated and used for analysis.

2.4.4. Perceived Stress Scale (PSS-10)

The 10-item Perceived Stress Scale is a shortened version of the original 14-item scale (PSS-14) developed to measure perceived stress [43]. It assesses the extent to which individuals view their lives as uncontrollable, unpredictable, and overwhelming over the previous month. The PSS-10 uses a 5-point Likert-type scale, ranging from 0 (Never) to 4 (Very Often). The total score ranges from 0 to 40, whereby scores between 0 to 13 are

considered low stress, scores between 14 to 26 are considered moderate stress, and scores between 27 to 40 are considered high stress.

2.4.5. *State-Trait Anxiety Inventory (STAI) and Visual Analog Scale (VAS)*

This study used the State Anxiety Subscale of the original inventory. The subscale contains 20 items developed to measure the participants' current feelings of anxiety. Each item is rated on a 4-point Likert scale, with a higher score indicating greater anxiety. The Visual Analog Scale (VAS) was used to measure the current level of stress on a scale from 0 to 10, with higher scores indicating higher levels of perceived stress.

2.4.6. *Cognitive Tests*

Cognitive tests are used to measure different aspects of a person's cognitive abilities, including memory, attention, problem-solving, processing speed, and reasoning. To induce mental fatigue, participants completed four cognitive tests consecutively, including N-back 2, Stroop test, go/no-go, and digitspan test. These tests were administered using PsyToolkit, a free online platform for cognitive and psychological experiments.

2.5. **Data Analysis**

All statistical analyses were performed in IBM SPSS Statistics (version 29.0). The Shapiro-Wilk test and Q-Q plot were used to test normal distribution. To compare the data before and after the treatment, a paired sample t-test was applied for normally distributed data, while the Wilcoxon signed-rank test was applied for non-normally distributed data, under the assumption of no carryover effects [44]. A significance level of $p < 0.05$ was considered for all statistical tests.

2.6. Results

The baseline characteristics of the participants are presented in Table 2.1. The study included 20 participants (9 male, 11 female) with moderate to high stress levels based on PSS. The mean age of the participants was 25.85 ± 4.09 , and the mean BMI was 23.87 ± 4.67 . 70% were single, 65% were postgraduate students, 90% were non-smokers, 50% exercised 3 to 4 times per week, and 65% reported sleeping six to seven hours per night.

Table 2.1: Demographic characteristics of the study participants

Characteristic	n = 20
Age, years, mean $\pm$ SD	25.85 ± 4.09
Weight, kg, mean $\pm$ SD	65.6 ± 17.03
Height, cm, mean $\pm$ SD	165.05 ± 8.07
BMI, kg/m ² , mean $\pm$ SD	23.87 ± 4.67
PSS, mean $\pm$ SD	22.8 ± 4.32
Gender, n (%)	
Male	9 (45)
Female	11 (55)
Marital status, n (%)	
Married	6 (30)
Single	14 (70)
Education, n (%)	
Undergraduate	7 (35)
Postgraduate	13 (65)
Smoking n (%)	
No	18 (90)
Yes	2 (10)
Exercise n (%)	
More than 4 times per week	2 (10)
3 to 4 times per week	10 (50)
1 to 2 times per week	4 (20)
None	4 (20)
Sleep duration n (%)	
More than 8 hours	1 (5)
7 to 8 hours	4 (20)
6 to 7 hours	13 (65)
Less than 6 hours	2 (10)

BMI: Body Mass Index; PSS: Perceived Stress Scale; SD: Standard deviation; n: number

2.6.1. Mental Fatigue Index

Table 2.2 presents the mental fatigue index at three time points: pre-cognitive test, post-cognitive test, and post-treatment. The cognitive test significantly increased the mental fatigue index in the massage condition from 12.88 ± 9.76 to 14.86 ± 11.11 ($p = 0.003$), while the increase from 13.78 ± 10.40 to 14.72 ± 11.19 in the control condition was not statistically significant ($p = 0.203$). The mental fatigue index decreased non-significantly after massage treatment from 14.86 ± 11.11 to 12.89 ± 9.68 ($p = 0.054$), but the decrease in mental fatigue index from 14.72 ± 11.19 to 12.60 ± 10.00 after the control treatment was significant ($p = 0.016$). No significant between-treatment differences were observed in mental fatigue changes from pre-cognitive test to post-cognitive test or from post-cognitive test to post-treatment ($p = 0.208$, 0.881). Furthermore, direct comparisons of the mental fatigue index between treatments found no significant differences between massage and control conditions at any time point: pre-cognitive test ($p = 0.166$), post-cognitive test ($p = 0.882$), or post-treatment ($p = 0.714$).

Table 2.2: Analysis of mental fatigue index pre-cognitive test, post-cognitive test, and post-treatment

	Massage (n = 20)	Control (n = 20)	p ²
Mental fatigue index			
T0	12.88 ± 9.76	13.78 ± 10.40	0.166 ^a
T1	14.86 ± 11.11	14.72 ± 11.19	0.882 ^a
T2	12.89 ± 9.68	12.60 ± 10.00	0.714 ^a
T1-T0	0.95 ± 3.21	1.99 ± 2.68	0.208 ^a
T1-T2	1.97 ± 4.29	2.13 ± 3.61	0.881 ^a
p ¹ (T1-T0)	0.003^b	0.203 ^a	
p ¹ (T1-T2)	0.054 ^a	0.016^a	

Values are presented as mean ± standard deviation.

T0: pre-cognitive test; T1: post-cognitive test; T2: post-treatment.

p < 0.05 are bold, p¹: within-treatment p-value; p²: between-treatment p-value.

^a paired sample t-test, ^b Wilcoxon signed-rank test.

2.6.2. Blood Pressure, Heart Rate, STAI, and VAS.

Table 2.3 presents the measurements for systolic blood pressure, diastolic blood pressure, heart rate, STAI, and VAS before and after treatment. The systolic blood pressure showed no significant difference after massage and control treatment (p = 0.140, 0.053). The diastolic blood pressure significantly decreased after massage treatment (p < 0.001), while no significant difference was found after control treatment (p = 0.548). The heart rate decreased significantly after massage and control treatment (p = 0.001, 0.021). The STAI and VAS scores significantly decreased after massage treatment (p < 0.001), but no significant differences were observed after control treatment (p = 0.184, 0.112). Between-treatment analysis showed significant differences for diastolic blood pressure, STAI, and VAS (p = 0.002, 0.008, 0.006), favouring massage treatment.

Table 2.3: Analysis of SBP, DBP, HR, STAI, and VAS before and after treatment

Variables	Massage (n = 20)		p ¹	ES	Control (n = 20)		p ¹	ES	p ²
	pre	post			pre	post			
SBP	102.82 ± 11.88	101.28 ± 10.93	0.140 ^a	0.345 ^c	104.85 ± 11.80	102.47 ± 11.35	0.053 ^a	0.462 ^c	0.617 ^a
DBP	70.03 ± 7.29	65.97 ± 6.54	<0.001^a	1.051 ^c	67.37 ± 7.45	66.85 ± 7.74	0.548 ^a	0.137 ^c	0.002^a
HR	73.25 ± 9.56	70.10 ± 9.53	0.001^a	0.856 ^c	73.43 ± 7.40	71.38 ± 9.04	0.021^a	0.560 ^c	0.391 ^a
STAI	40.95 ± 10.28	31.10 ± 6.74	<0.001^a	1.114 ^c	40.50 ± 9.59	37.85 ± 8.15	0.184 ^a	0.308 ^c	0.008^a
VAS	5.20 ± 2.14	3.25 ± 2.20	<0.001^a	1.363 ^c	4.70 ± 2.00	4.20 ± 2.22	0.112 ^b	0.502 ^d	0.003^a

Values are presented as mean ± standard deviation.

SBP: Systolic blood pressure; DBP: Diastolic blood pressure; HR: Heart rate; STAI: State-Trait Anxiety Inventory; VAS: Visual Analog scale; ES: Effect size.

p < 0.05 are bold, p¹: within-treatment changes before and after treatment; p²: between-treatment difference of the changes before and after treatment.

^a paired sample t-test.

^b Wilcoxon signed-rank test.

^c Cohen's d effect size.

^d Pearson's r effect size.

2.7. Discussion

Existing research on the effects of automated massage chair therapy remains limited. To the best of our knowledge, this is the first study to examine the effects of automated massage chair therapy on stress, anxiety, and mental fatigue among university students. Prior research has shown that perceived stress can contribute to subjective cognitive decline during crises [45]. Therefore, this study was focused on university students experiencing moderate to high stress levels, a condition known to worsen cognitive strain and reduce mental resilience. To induce mental fatigue among students, multiple cognitive tests are administered consecutively to the participants. Evidence from a previous study has shown that mental arithmetic tasks can successfully induce mental fatigue, and the brain activity during the task state differs significantly from that observed during the resting state [46]. In this study, EEG-based mental fatigue was used instead of self-reported mental fatigue, as it provides objective, real-time data on brain activity and is considered more sensitive to mental fatigue than subject self-reporting data [47]. In another study, different algorithms were used to assess fatigue levels in sleep-deprived drivers after

performing a monotonous driving task, and the increase in $(\theta + \alpha)/\beta$ was greater compared to the other algorithms [48]. Therefore, in this study, the mental fatigue index was calculated using the algorithm $(\theta + \alpha)/\beta$ and assessed across three distinct time points: before cognitive tests, after cognitive tests, and after massage treatment. The results showed that the cognitive test significantly increased the mental fatigue index in the massage condition, while the increase in the control condition was not statistically significant. This may be due to a higher baseline mental fatigue level in the control condition, potentially limiting the increase in mental fatigue. The baseline difference, while not statistically significant, may reflect random variation due to factors such as sleep quality or stress levels before the study. No significant differences in the mental fatigue levels were observed between the intervention and control conditions after the cognitive tests, suggesting that the cognitive tests induced a similar level of mental fatigue. Interestingly, while the control condition showed a statistically significant decrease in mental fatigue after treatment, the mental fatigue reduction in the massage condition was not significant ($p = 0.054$). However, when comparing the changes between the two conditions, no statistically significant difference in mental fatigue reduction was found. This suggests that both the massage and control treatment may have contributed to reduced mental fatigue, but neither demonstrated a superior effect in this study. In contrast, Lim, Kim [49] reported no significant reduction in the mental fatigue index with mechanical massage or control treatment alone, while mechanical massage combined with binaural beats significantly reduced mental fatigue. In fact, their findings showed that massage paired with binaural beats was significantly more effective at reducing mental fatigue than massage or control treatments alone. One possible explanation is that the mental fatigue induced in this study may not have been severe enough to require any intervention. It is possible that the mild mental fatigue experienced by participants was alleviated simply through passive rest in the massage chair. Therefore, future studies should use different mental fatigue protocols to induce higher mental fatigue.

Blood pressure and heart rate tend to vary throughout the day due to influences such as physical activity, stress, and the body's natural circadian rhythm. Massage therapy has been shown to reduce the levels of stress hormones such as cortisol and adrenaline, while promoting the release of feel-good hormones like oxytocin [50, 51]. Research showed that higher oxytocin levels are linked to lower heart rate and blood pressure in premenopausal women [52]. Therefore, in this study, changes in blood pressure and heart rate were used as objective indicators for stress and anxiety. The results of this study show that massage treatment significantly reduces diastolic blood pressure, while the decrease in systolic blood pressure was not significant after massage. Previous studies studying the effects of massage therapy on stress in healthy individuals have consistently reported a statistically significant reduction in both systolic and diastolic blood pressure [53, 54]. In some studies, the statistically significant reduction is observed only for systolic blood pressure [55, 56]. In contrast, the present study reported a statistically significant decrease in diastolic blood pressure alone. This suggests that massage therapy may have a more pronounced effect on vascular tone rather than cardiac output, as increased vascular tone results in elevated diastolic pressure [57]. An additional explanation could be the delay in post-treatment blood pressure measurement due to the preceding EEG recording. The three-minute delay may have allowed the transient effects of automated massage on systolic blood pressure to diminish, as systolic blood pressure is more sensitive to small changes in stress level and physical activity. The diastolic blood pressure likely remained reduced due to the longer-lasting effects of decreased vascular tone. In addition to changes in blood pressure, this study also examined heart rate as another indicator of stress and anxiety. The heart rate reduced significantly after both massage and control treatment, and there is no significant difference in the heart rate changes between both treatments. This result is consistent with the previous study by Olney [58], which reported no significant difference between back massage and control treatment on heart rate changes. These results suggest that resting alone may be enough to reduce the heart rate.

In this study, self-reported measures like the STAI and VAS were used to evaluate how participants felt after the treatment because stress and anxiety are inherently subjective and complex experiences influenced by many factors. The results show that stress and anxiety reduced significantly after massage treatment compared to the control treatment. Therefore, these findings are consistent with the results from blood pressure and heart rate, further emphasizing the effect of massage therapy in reducing stress and anxiety levels among moderately stressed students. In fact, a previous study involving high school students who smoked reported that self-hand massage resulted in significant improvement in stress levels in the experimental group compared to the control group [59]. The alignment between the current study and the previous study suggests that both self-administered massage and automated massage may trigger physiological and psychological responses that help reduce stress and anxiety. One possible explanation revolves around mental disengagement during massage therapy, which allows the

participant to temporarily forget about the stress they had. The act of resting on the massage chair to receive the massage helps the participant to calm down and remain present in the moment. This psychological mechanism, in combination with other physiological mechanisms, might have resulted in significant stress and anxiety reduction after massage therapy.

3. Part 2: Effects of Automated Massage Chair Therapy on Blood Pressure, Heart Rate, and Skin Blood Flow

3.1. Methodology

3.1.1. Study design

This one-group pretest-posttest pilot study aimed to evaluate the immediate effects of automated massage chair therapy on blood pressure, heart rate, and skin blood flow for hypertensive hospital staff at University Malaya Medical Centre (UMMC). Each participant received three sessions of automated massage chair therapy, with a 48-hour washout period between sessions to minimize carryover effects. The study took place at the Clinic of Rehabilitation Robotics, Exercise and Advanced Technology (CREATE), Menara Selatan, UMMC.

3.1.2. Participants and Recruitment

An invitation email was distributed via the University Malaya Medical Centre mailing list to recruit participants through convenience sampling. The participants were fully informed about the study procedures, as well as the potential benefits and risks of the intervention. Participation in the study was completely voluntary, and participants were free to withdraw at any time without consequence. All participants provided informed written consent before data collection. The eligibility criteria were as follows: 1) age between 18-60 years old, 2) systolic blood pressure of 120-140 mmHg or diastolic blood pressure of 80-90 mmHg, 3) able to participate for the entire study duration, 4) not pregnant, 5) not receiving concurrent massage therapy, 6) no active psychiatric condition or neurological condition. The study was reviewed and approved by the Medical Research Ethics Committee (MREC) of University Malaya Medical Centre (Identification No. 20231122-13049).

3.1.3. Experimental Procedure and Intervention

The overall experimental procedure for this study is shown in Figure 3.1. Every participant received three sessions of automated massage chair therapy on alternate days to ensure adequate spacing and minimize carryover effects. Before every session, participants were seated on the massage chair for 10 minutes to acclimatize to the settings. Baseline assessments began with blood pressure measurement, followed by skin blood flow measurement. Participants then received a 20-minute automated massage chair therapy with the deep tissue massage program. Immediately after the treatment, the same measurements during the baseline assessment were used; however, skin blood flow measurement took precedence. The participants were instructed to remain still and avoid falling asleep during the experiment. They were also asked to refrain from consuming caffeine, alcohol, or other stimulants for at least 12 hours before the session. The demographic characteristics of the study participants were collected using a questionnaire on their first visit.

3.2. *Equipment and Outcome Measures*

3.2.1. *Ogawa Master Drive AI 2.0 (OG-7598X) Massage Chair*

The automated massage chair therapy was delivered using Ogawa Master Drive AI 2.0 (OG7598X, Ogawa, Malaysia). The massage chair's pre-programmed massage mode (deep tissue massage program) was used to massage the whole body for 20 minutes.

3.2.2. *Blood Pressure and Heart Rate*

Blood pressure and heart rate were measured using an Omron automatic blood pressure monitor HEM-7156T (Omron, Japan).

3.2.3. *Skin Blood Flow with Laser Doppler Flowmeter*

Skin blood flow was measured with Periflux System 5000 (Perimed, Sweden), a multi-channel instrument for microvascular assessment. This system can be configured with up to four function units, including PF 5010, PF 5020, PF 5040, and PF 5050. In this study, PF 5010 Laser Doppler Perfusion Monitoring unit was used to measure skin blood flow in arbitrary perfusion units (PU). To minimize the influence of room temperature on skin perfusion, PF 5020 Temp Unit was used to heat the skin tissue to 32 °C. Probe 457 is used to provide local heating while simultaneously measuring skin blood flow. The form factor of Probe 457 is shown in Figure 3.2a. Initially, the participant sat on the massage chair with their left forearm slightly abducted and resting on the automated massage chair armrest. Before measurements began, the superficial dermis of the antecubital fossa region was first cleaned with an alcohol pad. The optical fiber probe was then attached to the skin using double-sided tape, as shown in Figure 3.2b. The skin blood flow was recorded continuously for 5 minutes.

3.2.4. *Data analysis*

IBM SPSS Statistics (version 29.0) was used for data analysis. All data were tested for normality using the Shapiro-Wilk test. Different data analysis methods were applied to blood pressure and skin blood flow data. For skin blood flow, the intrasession repeatability of the laser Doppler flowmeter was assessed using the intraclass correlation coefficient (ICC), calculated through a two-way mixed effects model with absolute agreement. The mean absolute difference between pre-intervention measurements for different sessions was calculated to evaluate the intersession repeatability. Bland-Altman plots were constructed to examine the intervention effect and individual variability in response to the treatment. Friedman's test was used to detect differences in medians across all six repeated measurements. For blood pressure, repeated measures ANOVA were used to examine the effects of time, session, and time*session interaction. Paired sample t-tests were then applied to compare blood pressure before and after the treatment for each session. A significance level of $p < 0.05$ was considered for all statistical tests.

3.3. Results

The baseline characteristics of the participants are presented in Table 3.1. The study included 20 participants (5 males, 15 females) with hypertension. The mean age was 45.0 ± 8.07 , mean weight 71.08 ± 14.05 , mean height 158.8 ± 9.37 , and mean BMI 28.24 ± 5.28 . Most participants were married (85%), reported moderate stress levels (80%), were non-smokers (100%), and did not consume alcohol (95%). The majority were receiving antihypertensive medication (85%) and engaged in physical exercise at least once per week (80%).

Table 3.1: Demographic characteristics of the study participants

Characteristic	n = 20
Age, years, mean $\pm$ SD	45.0 $\pm$ 8.07
Weight, kg, mean $\pm$ SD	71.08 $\pm$ 14.05
Height, cm, mean $\pm$ SD	158.8 $\pm$ 9.37
BMI, kg/m ² , mean $\pm$ SD	28.24 $\pm$ 5.28
Gender, n (%)	
Male	5 (25)
Female	15 (75)
Marital status, n (%)	
Married	17 (85)
Single	3 (15)
Stress, n (%)	
Low	3 (15)
Moderate	16 (80)
High	1 (5)
Smoking, n (%)	
Yes	0 (0)
No	20 (100)
Alcohol	
Yes	1 (5)
No	19 (95)
Hypertensive medication, n (%)	
Yes	17 (85)
No	3 (15)
Exercise, n (%)	
More than 2 times per week	6 (30)
1 to 2 times per week	10 (50)
None	4 (20)

BMI: Body Mass Index; SD: Standard deviation; n: number

3.3.1. Skin Blood Flow

Table 3.2 shows the ICC values for skin blood flow measurements calculated twice each session (pre-test and post-test) to assess the intrasession reliability of the measurements. The intrasession reliability of skin blood flow measurements are all above 0.99.

Table 3.2: Intraclass correlation coefficient (ICC) for the skin blood flow measurement at different timepoint (N = 19)

Session	Time	ICC
Session 1	Pre	0.999
	Post	0.999
Session 2	Pre	0.998
	Post	0.997
Session 3	Pre	0.998
	Post	0.994

Table 3.3 shows the mean absolute difference calculated using pre-session data from two different sessions to evaluate the intersession repeatability of skin blood flow measurements. The results showed a high mean absolute difference and a wide confidence interval.

Table 3.3: Intersession repeatability for skin blood flow measurement (N = 19)

Variable	Session 1 & Session 2	Session 1 & Session 3	Session 2 & Session 3
Mean absolute difference	10.40	8.03	10.07
95% confidence interval	4.99 to 15.81	3.49 to 12.56	5.05 to 15.10

Figures 3.3, 3.4, and 3.5 present Bland-Altman plots to evaluate the intervention effect and individual variability in response to the treatment. The plots indicated that the intervention effect varies across different sessions, and the wide limits of agreement in all three plots indicate substantial individual variability in response to the treatment. In addition, a trend is observed in Bland-Altman plots, whereby the differences increase with the magnitude of the measurements. After evaluating the variability of skin blood flow measurements within and between sessions, Friedman's test was used to detect differences in ranks across all six repeated measurements since the normality test indicated that the data were not normally distributed. The results showed that there was no significant difference in ranks across the repeated measurements, $\chi^2(5) = 1.135$, $p = 0.951$. This suggests that the overall distributions of the repeated measurements are the same.

3.3.2. Blood Pressure and Heart Rate

Table 3.4 presents the result of the repeated measures ANOVA examining the effects of time, session, and their interaction (time*session) on systolic blood pressure, diastolic blood pressure, and heart rate. The results revealed a significant main effect of time and a significant main effect of session for all three variables ($p < 0.05$), but no significant interaction (time*session) effect.

Table 3.4: Repeated measures ANOVA for systolic blood pressure, diastolic blood pressure and heart rate (n = 20)

Effect	Measurement	df	F	p
time (pre, post)	systolic blood pressure	(1,19)	16.506	< 0.001
	diastolic blood pressure	(1,19)	5.282	0.033
	heart rate	(1,19)	20.558	< 0.001
session (s1, s2, s3)	systolic blood pressure	(2,38)	5.045	0.011
	diastolic blood pressure	(2,38)	8.787	< 0.001
	heart rate	(2,38)	4.664	0.015
time*session	systolic blood pressure	(2,38)	0.701	0.701
	diastolic blood pressure	(2,38)	1.769	0.184
	heart rate	(2,38)	0.118	0.889

df: degree of freedom; s1: session 1; s2: session 2; s3: session 3

$p < 0.05$ are bold

Tables 3.5, 3.6, and 3.7 present the results of the paired sample t-test on systolic blood pressure, diastolic blood pressure, and heart rate. The systolic blood pressure, diastolic blood pressure, and heart rate decrease significantly after massage therapy in all sessions ($p < 0.05$), except for systolic and diastolic blood pressure in session 2. Cohen's d effect size is computed to understand the magnitude of changes. A Cohen's d of 0.5 indicates a medium-sized difference.

Table 3.5: Paired sample t-test comparing systolic blood pressure before and after massage therapy (N=20)

Session	Time	Mean $\pm$ SD	MD	t	df	p	Cohen's d
Session 1	Pre	124.97 $\pm$ 14.07	-3.43	2.901	19	0.009	0.649
	Post	121.53 $\pm$ 13.48					
Session 2	Pre	120.05 $\pm$ 12.61	-2.32	1.895	19	0.073	0.424
	Post	117.73 $\pm$ 11.96					
Session 3	Pre	119.50 $\pm$ 10.90	-3.54	2.976	19	0.008	0.665
	Post	115.96 $\pm$ 12.16					

SD: standard deviation; MD: mean difference; df: degrees of freedom

$p < 0.05$ are bold

Table 3.6: Paired sample t-test comparing diastolic blood pressure before and after massage therapy (N=20)

Session	Time	Mean $\pm$ SD	MD	t	df	p	Cohen's d
Session 1	Pre	83.25 $\pm$ 10.56	-2.25	2.374	19	0.028	0.531
	Post	81.00 $\pm$ 9.83					
Session 2	Pre	77.38 $\pm$ 10.43	0	0	19	1.000	0
	Post	77.38 $\pm$ 9.57					
Session 3	Pre	78.23 $\pm$ 8.87	-1.73	2.165	19	0.043	0.484
	Post	76.50 $\pm$ 9.28					

SD: standard deviation; MD: mean difference; df: degrees of freedom

$p < 0.05$ are bold

Table 3.7: Paired sample t-test comparing heart rate before and after massage therapy (N=20)

Session	Time	Mean $\pm$ SD	MD	t	df	p	Cohen's d
Session 1	Pre	74.78 $\pm$ 12.04	-4.07	3.455	19	0.003	0.773
	Post	70.72 $\pm$ 10.88					
Session 2	Pre	74.85 $\pm$ 7.80	-3.85	2.349	19	0.030	0.525
	Post	71.00 $\pm$ 9.58					
Session 3	Pre	77.82 $\pm$ 11.11	-3.22	2.871	19	0.010	0.642
	Post	74.60 $\pm$ 9.11					

SD: standard deviation; MD: mean difference; df: degrees of freedom

p < 0.05 are bold

3.4. Discussion

In this study, skin blood flow data for one participant were unavailable due to data acquisition issues. The final analysis, therefore, included 19 participants to determine the reliability of skin blood flow measurement. Intrasection reliability was assessed using the intraclass correlation coefficient (ICC). ICC was calculated with a two-way mixed-effects model based on absolute agreement, using 20 consecutive measurements (taken immediately before and after treatment) for each participant at every measurement session. The result showed that all the ICCs are above 0.99. This suggests that, within a single session, the skin blood flow measurements are highly consistent. However, it is important to note that these results fail to account for intersession repeatability. To evaluate the intersession repeatability of tissue perfusion measurement with a laser Doppler flowmeter, the mean absolute difference is calculated using pre-session data from two different sessions. The results show a high mean absolute difference and a wide confidence interval, indicating a high variability for the measurement between different sessions. These findings suggest that different measuring conditions or individual biological fluctuation might have resulted in the variability when comparing measurements across different sessions.

The Bland-Altman plot is used to evaluate the intervention effect and individual variability in response to the treatment. The middle line on the Bland-Altman plot represents the average intervention effect, while the upper and lower lines represent the 95% limits of agreement where most individual differences are expected to fall. The plot showed that there is a high individual variability, and the intervention effect differed across sessions. There could be several explanations for the high individual variability in response to the treatment. First, the laser Doppler flowmeter could have contributed to reliability issues, as it uses a single-point probe that samples a relatively small tissue area and is highly sensitive to movement artifacts [60]. Next, there could be an inconsistency in the placement of the probe after massage therapy. In this study, a probe holder was not used to secure the probe; therefore, a slight displacement of the probe after massage therapy could have resulted in increased variability. Finally, results from Friedman's test showed no significant difference in ranks between repeated measurements, suggesting no changes in skin blood flow.

The present study demonstrates that systolic blood pressure, diastolic blood pressure, and heart rate reduce significantly after automated massage chair therapy with a significant main effect of time. These findings align with previous studies supporting the use of massage therapy as an effective non-pharmacological intervention for managing hypertension. For example, Moghadasi, Mohebbi [61] demonstrate that six sessions of Swedish massage significantly reduced blood pressure and heart rate in hypertensive patients compared to a control group. A significant main effect of session was also observed, indicating differences in blood pressure and heart rate across the three sessions. However, there is no significant interaction (time*session), suggesting that the change from pre- to post-measurements is consistent across all sessions. This implies that automated massage chair therapy had an equal effect on blood pressure and heart rate in all sessions. Therefore, the significant main effect of session is likely due to the difference in baseline, rather than differences in the effectiveness of the intervention across sessions. The observed reduction in blood pressure and heart rate may result from a combination of physiological mechanisms. Deep tissue massage is effective in reaching muscles and soft tissues deep under the skin, helping to relieve muscle tension. The relaxation of tight muscles reduces pressure on the surrounding blood vessels, allowing vasodilation and lowering of blood pressure [62]. In addition, massage therapy may stimulate the release of endothelial nitric oxide. The increase in blood flow resulted in a temporary rise in shear stress on the vessel walls, prompting the release of endothelial nitric oxide. This results in the relaxation and widening of the blood vessels, improving blood flow and ultimately reducing blood pressure [63].

4. Part 3: Effects of Automated Massage Chair Therapy on Low Back Function and Pain

4.1. Methodology

4.1.1. Study Design

This one-group pretest-posttest pilot study aimed to evaluate the immediate effects of automated massage chair therapy on low back function and pain for chronic low back pain hospital staff at University Malaya Medical Centre (UMMC). Each participant received three sessions of automated massage chair therapy, with a 48-hour washout period between sessions to minimize carryover effects. The study took place at the Clinic of Rehabilitation Robotics, Exercise and Advanced Technology (CREATE), Menara Selatan, UMMC.

4.1.2. Participants and Recruitment

An invitation email was distributed via the University Malaya Medical Centre mailing list to recruit participants through convenience sampling. The participants were fully informed about the study procedures, as well as the potential benefits and risks of the intervention. Participation in the study was completely voluntary, and participants were free to withdraw at any time without consequence. All participants provided informed written consent before data collection. The eligibility criteria were as follows: 1) age between 18-60 years old, 2) chronic low back pain for more than 12 weeks, 3) able to participate for the entire study duration, 4) not pregnant, 5) not receiving concurrent massage therapy, 6) no active psychiatric condition or neurological condition. The study was reviewed and approved by the Medical Research Ethics Committee (MREC) of University Malaya Medical Centre (Identification No. 20231122-13049).

4.1.3. Experimental Procedure and Intervention

The overall experimental procedure for this study is shown in Figure 4.1. Every participant received three sessions of automated massage chair therapy on alternate days to ensure adequate spacing and minimize carryover effects. Before every session, participants were seated on the massage chair for 10 minutes to acclimatize to the environment. Baseline assessments began with self-reported low back pain using the Visual Analogue Scale (VAS), followed by electromyogram (EMG) recording of the erector spinae muscle while participants performed three sets of trunk flexion and extension. Participants then received a 20-minute automated massage chair therapy with the spinal release pressure massage program. Immediately after the treatment, the same assessments conducted at baseline were repeated; however, electromyogram recording took precedence. Participants were instructed to remain still and avoid falling asleep during the experiment. They were also asked to refrain from consuming caffeine, alcohol, or other stimulants for at least 12 hours before each session and to ensure adequate sleep the night before. The demographic characteristics of the study participants were collected using a questionnaire on their first visit.

4.2. Equipment and Outcome Measures

4.2.1. Ogawa Master Drive AI 2.0 (OG-7598X) Massage Chair

The automated massage chair therapy was delivered using Ogawa Master Drive AI 2.0 (OG7598X, Ogawa, Malaysia). The massage chair's pre-programmed massage mode (spinal release pressure program) was used to massage the whole body for 20 minutes.

4.2.2. Electromyogram

Muscle activity was acquired using Delsys Trigno Wireless EMG System (Delsys, Natick, MA, USA). EMG data were collected from the erector spinae muscle bilaterally using sensors placed 3 cm lateral to the spinous processes of L3/L4. The sensors were positioned vertically along the spine. Before data acquisition began, the participants were instructed to perform trunk flexion as a baseline test to ensure proper signal quality. After checking the signal quality, the data recording will start. The participants were instructed to flex their trunk slowly over 5 seconds as far as possible, hold the maximum bent position for 2 seconds, and return to the upright position slowly over 5 seconds. MATLAB R2024b was used for data preprocessing, including filtering, root mean square (RMS)

calculation, and extension relaxation ratio calculation. The data were collected at a sampling rate of 2000 Hz. The recordings were then processed offline using a 20 Hz high-pass filter and 450 Hz low-pass filter, while the 50 Hz line interference was filtered using a notch filter. The bandpass filter and notch filter were implemented using a fourth-order Butterworth filter. The data was further processed by calculating the RMS value with a 100-ms sliding window. To assess abnormal muscle activation patterns commonly associated with low back pain, the extension relaxation ratio (ERR) was computed. ERR was calculated as the ratio of the maximum one-second averaged RMS EMG value during the trunk extension phase to the maximum one-second averaged RMS EMG value during the full flexion phase.

4.2.3. Visual Analog Scale

The Visual Analog Scale (VAS) was used to measure the participants' low back pain on a 0 - 10 scale, where a higher score indicates higher pain.

4.2.4. Data Analysis

All statistical analyses were performed in IBM SPSS Statistics (version 29.0). Different data analysis methods were applied to EMG and VAS data. The Shapiro-Wilk test was used to test normal distribution. For EMG data, a linear mixed-effects model was used to examine the effects of time, session, and their interaction (time*session). The Wilcoxon signed-rank test was used to compare EMG before and after treatment in each session. For VAS, Friedman's test was used to detect differences in medians across all repeated measurements, while the Wilcoxon Signed-Rank test was used to compare VAS before and after treatment in each session. A significance level of $p < 0.05$ was considered for all statistical tests.

4.3. Results

The baseline characteristics of the participants are presented in Table 4.1. The study included 20 participants (5 males, 15 females) with chronic low back pain. The mean age was 34.65 ± 9.40 , mean weight 70.1 ± 14.7 , mean height 161.05 ± 8.06 , and mean BMI 26.87 ± 4.1 . Most participants were married (70%), reported mild low back pain (55%), and took low back pain medications (90%).

Table 4.1: Demographic characteristic of the study participants

Characteristic	n = 20
Age, years, mean $\pm$ SD	34.65 ± 9.40
Weight, kg, mean $\pm$ SD	70.1 ± 14.7
Height, cm, mean $\pm$ SD	161.05 ± 8.06
BMI, kg/m ² , mean $\pm$ SD	26.87 ± 4.1
Gender, n (%)	
Male	5 (25)
Female	15 (75)
Marital status, n (%)	
Married	14 (70)
Single	6 (30)
Low back pain, n (%)	
Mild	11 (55)
Moderate	9 (45)
Severe	0 (0)
Low back pain medication, n (%)	
No	18 (90)
Yes	2 (10)
Low back pain trigger, n (%)	
Bending or twisting	1 (5)
Lifting heavy objects	6 (30)
Sitting for long periods	7 (35)
Standing for long periods	6 (30)

BMI: Body Mass Index; SD: Standard deviation; n: number

4.3.1. Extension Relaxation Ratio

Table 4.2 presents the results of the linear mixed-effects model examining the effects of time, session, and their interaction (time*session) on left and right ERR. The result revealed a significant main effect of time for left ERR

($p < 0.001$), whereas no significant main effect of time was observed for right ERR ($p = 0.068$). In addition, neither session nor the time*session interaction showed significant effects ($p > 0.05$).

Table 4.2: Linear mixed-effect model for the left and right extension relaxation ratio ($n = 20$)

Effect	Measurement	df	F	p
time (pre, post)	left ERR	(1,87.088)	12.812	<0.001
	right ERR	(1,89.093)	3.410	0.068
session (S1,S2,S3)	left ERR	(2,87.717)	2.891	0.061
	right ERR	(2,89.422)	1.200	0.306
time*session	left ERR	(2,87.088)	1.969	0.146
	right ERR	(2,89.093)	2.255	0.111

S1: session 1; S2: session 2; S3: session 3.

ERR: extension relaxation ratio; df: degrees of freedom

$p < 0.05$ are bold

Tables 4.3 and 4.4 present the results of the Wilcoxon signed rank tests on left ERR and right ERR. The left ERR and right ERR increase significantly after automated massage chair therapy in session 1 ($p = 0.006$). No significant changes were observed for session 2 and session 3 ($p > 0.05$).

Table 4.3: Wilcoxon signed-rank test comparing left ERR before and after massage therapy ($n = 20$)

Session	Time	mean $\pm$ SD	MD	Z	p
Session 1	Pre	5.44 $\pm$ 3.30	2.39	-2.722	0.006
	Post	7.83 $\pm$ 4.78			
Session 2	Pre	5.11 $\pm$ 2.54	0.90	-1.083	0.279
	Post	6.02 $\pm$ 4.53			
Session 3	Pre	5.00 $\pm$ 2.52	0.69	-1.408	0.159
	Post	5.69 $\pm$ 2.57			

ERR: extension relaxation ratio; SD: standard deviation; MD: mean difference

$p < 0.05$ are bold

Table 4.4: Wilcoxon signed-rank test comparing the right ERR before and after massage therapy ($n = 20$)

Session	Time	mean $\pm$ SD	MD	Z	p
Session 1	Pre	4.61 $\pm$ 3.98	1.56	-2.772	0.006
	Post	6.16 $\pm$ 4.37			
Session 2	Pre	4.42 $\pm$ 3.33	0.56	-0.597	0.550
	Post	4.99 $\pm$ 3.61			
Session 3	Pre	4.85 $\pm$ 4.50	-0.24	-0.604	0.546
	Post	4.61 $\pm$ 2.73			

ERR: extension relaxation ratio; SD: standard deviation; MD: mean difference

$p < 0.05$ are bold

4.3.2. Visual Analog Scale

Friedman's test indicated a significant difference in repeated measurements of low back pain, as measured with VAS ($p < 0.001$). Therefore, post-hoc analysis was carried out. Table 4.5 presents the results of the Wilcoxon signed rank test on VAS. The results showed VAS decreased significantly after automated massage chair therapy for all three sessions ($p < 0.001$).

Table 4.5: Wilcoxon signed-rank test comparing low back pain (VAS) before and after massage therapy ($n = 20$)

Session	Time	mean $\pm$ SD	MD	Z	p
Session 1	Pre	5.10 $\pm$ 1.41	-1.15	-3.906	<0.001
	Post	3.95 $\pm$ 1.47			
Session 2	Pre	4.70 $\pm$ 1.69	-1.45	-3.896	<0.001
	Post	3.25 $\pm$ 1.07			
Session 3	Pre	3.95 $\pm$ 1.28	-1.25	-4.017	<0.001
	Post	2.70 $\pm$ 1.03			

SD: standard deviation; MD: mean difference, $p < 0.05$ are bold

4.4. Discussion

The flexion relaxation phenomenon (FRP) is characterized by the myoelectric silence of the erector spinae muscle during full trunk flexion. In a healthy individual, this phenomenon reduces the activation of muscles and allows passive spinal structures to support the body. However, the flexion relaxation phenomenon is often absent or disrupted in individuals with low back pain, as the erector spinae muscle tends to remain active even during full

trunk flexion. This abnormal muscle activation may reflect protective guarding, increase spinal instability, or alter motor control strategies in response to pain or fear of movement. In this study, EMG activity collected during trunk flexion and extension movement was used to calculate the extension relaxation ratio (ERR). The ERR is calculated by dividing EMG activity during extension by the EMG activity during full flexion. A high ERR suggests that the flexion relaxation phenomenon may be present, as it reflects reduced muscle activity during full flexion compared to extension. In contrast, a low ERR may indicate dysfunction of the flexion relaxation phenomenon, which is commonly observed in individuals with low back pain. To objectively and quantitatively assess low back pain, the extension relaxation ratio (ERR) was calculated for both left and right erector spinae muscles, referred to as left ERR and right ERR, respectively.

In this study, a linear mixed-effects model was used to investigate changes in muscle activation of the erector spinae muscle. The result revealed a significant main effect of time for left ERR, but no significant main effect of time was observed for right ERR. In addition, neither session nor the time*session interaction showed significant effects. The lack of a significant effect for the time*session interaction suggests that the change from pre- to post-measurements is consistent across sessions. Likewise, the lack of a significant main effect of session implies that the baseline measurements were similar across sessions. Later, post-hoc analysis using the Wilcoxon signed-rank test showed that left ERR and right ERR increased significantly in session 1, whereas changes in session 2 and session 3 were not significant.

The overall increase in ERR suggests a restoration of the flexion relaxation phenomenon, which is typically absent or impaired in individuals with chronic low back pain. According to Weerapong, Hume [34], massage has been widely utilized to reduce muscle tension, alleviate muscle pain, and promote muscle relaxation. One proposed explanation for the increase in ERR is the reduction of low back pain after massage therapy. Pressure applied during massage therapy stimulates the sensory receptors in the skin, thereby triggering the release of endorphin from the pituitary gland [64]. Endorphins are inhibitory neurotransmitters that block pain signals, reducing the perception of pain. Pain, while serving as a protective mechanism, changes motor control patterns to prevent further damage. In the early stages of injury, this protective mechanism helps with healing, but if maintained for too long, it may lead to more pain in the affected structure or nearby structures, due to suboptimal muscle loading. By reducing pain through endorphin release, massage therapy may help reduce protective muscle guarding, restore normal motor control, and improve the flexion relaxation response in individuals with low back pain. Another possible explanation for the increase in ERR is reduced neuromuscular excitability, as reflected by decreased H-reflex amplitude on the muscle that receives the massage [65]. The H-reflex is an electrically evoked spinal reflex analogous to the mechanically evoked spinal stretch reflex, recorded as muscle activity from a muscle [66]. In individuals with low back pain, reduced neuromuscular excitability may help minimize the abnormal activation of the erector spinae muscle, thereby allowing them to relax fully during full trunk flexion. However, studies showed that H-reflex amplitude returns to baseline levels immediately after the massage ends [67]. Therefore, additional research is needed to better understand the mechanism through which massage therapy improves ERR in low back pain participants.

The perception of pain is a complex experience that cannot be measured objectively like muscle activity. Understanding the pain associated with an injury or disease is fundamental to evaluating its severity, determining appropriate intervention, and assessing the effectiveness of treatments. To directly measure the changes in perception of pain after massage therapy, VAS is administered to the participant before and after the automated massage chair therapy for all three sessions, and statistical analysis was carried out to evaluate the changes in VAS across repeated sessions.

The results showed that low back pain measured by VAS decreased significantly after automated massage chair therapy across all three sessions. In this study, the consistent reduction in VAS scores across all sessions not only supports the effectiveness of massage in alleviating low back pain but also reinforces the clinical relevance of observed restoration in the flexion relaxation phenomenon, as these physiological changes are reflected in the participant's perception of pain. In fact, these findings align with a previous research that reported massage therapy applied twice a week for six weeks alleviates low back pain [68]. Furthermore, massage therapy demonstrated a cumulative effect, as the mean VAS score steadily declined from 5.10 to 2.70 after three sessions of massage therapy. The cumulative effect suggests that massage therapy may be more beneficial when applied consistently over time to individuals with low back pain.

5. Limitations

The present study has some limitations that need to be considered. First, the small sample size ($n = 20$) may affect the generalizability of the findings. A small sample size increases the likelihood of outliers, which can skew the distribution and distort the results of the analysis. More participants are required to increase statistical power and improve the reliability and validity of the experimental findings. Next, the study mainly focuses on the immediate effects of automated massage chair therapy; long-term effects were not considered. In addition, this study is limited by the measurement protocol and the equipment used. The blood pressure measured in the presence of the researcher may have been confounded by the white coat effect. Therefore, further studies should consider using an automated blood pressure monitor that takes multiple readings while the participant is resting alone to eliminate the white coat effect [69].

6. Conclusion

The preliminary results from these pilot studies showed that automated massage chair therapy can reduce stress, mental fatigue, blood pressure, and low back pain among individuals with stress. However, these results should be interpreted with caution, as the sample size was small and the study was conducted in a controlled research setting. The intervention was easy to administer, required minimal supervision, and was well tolerated by participants. In fact, all participants completed the study without developing any side effects after the massage therapy.

7. Implications

Automated massage chair therapy offers several advantages as a complementary healthcare approach. Its ease of use, lack of required commitment, and suitability for older adults or individuals with mobility issues make it a practical and inclusive option. In the future, technological advances may enable AI-driven personalized massage programs tailored to individual's needs, thereby enhancing both effectiveness and user engagement. All in all, widespread implementation of such devices in public spaces including universities and hospitals could contribute to improved health equity, providing accessible support for both physical and mental health across diverse populations.

Glossary

BMI: Body Mass Index, CAM : Complementary and Alternative Medicine, SBP: Systolic Blood Pressure, DBP: Diastolic Blood Pressure, EEG: Electroencephalogram, ERR: Extension Relaxation Ratio, FFT: Fast Fourier Transform, GABA: Gamma-Aminobutyric Acid, HPA: Hypothalamic-Pituitary-Adrenal, ICA: Independent Component Analysis, ICC: Intraclass Correlation Coefficient, MVC: Maximal Voluntary Contraction, PSS: Perceived Stress Scale, RMS: Root Mean Squared, STAI: State-Trait Anxiety Inventory, VAS: Visual Analog Scale,

Funding

This work was supported by the Joint Research Grant, Universiti Malaya-Ogawa Malaysia, under Grant MG007-2023 and PV006-2023.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used CHATGPT to improve the clarity and structure of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication

References

1. Yi, D., et al. *Does Involution Cause Anxiety? An Empirical Study from Chinese Universities*. International Journal of Environmental Research and Public Health, 2022. **19**, DOI: 10.3390/ijerph19169826.
2. Liu, A.C., et al., *Influence of academic involution atmosphere on college students' stress response: the chain mediating effect of relative deprivation and academic involution*. BMC PUBLIC HEALTH, 2024. **24**(1).
3. Jiang, S.H., et al., *Regional Internet Access and Mental Stress Among University Students: A Representative Nationwide Study of China*. FRONTIERS IN PUBLIC HEALTH, 2022. **10**.
4. Ito, S., et al., *Occupational stress among healthcare workers in Japan*. WORK-A JOURNAL OF PREVENTION ASSESSMENT & REHABILITATION, 2014. **49**(2): p. 225–234.
5. Sharan, P. and C. Vellapandian, *Hypothalamic-Pituitary-Adrenal (HPA) Axis: Unveiling the Potential Mechanisms Involved in Stress-Induced Alzheimer's Disease and Depression*. CUREUS JOURNAL OF MEDICAL SCIENCE, 2024. **16**(8).
6. Degroote, C., et al., *Acute Stress Improves Concentration Performance Opposite Effects of Anxiety and Cortisol*. EXPERIMENTAL PSYCHOLOGY, 2020. **67**(2): p. 88–98.
7. Pérez-Jorge, D., et al., *Examining the effects of academic stress on student well-being in higher education*. HUMANITIES & SOCIAL SCIENCES COMMUNICATIONS, 2025. **12**(1).
8. Deng, Y.W., et al., *Family and Academic Stress and Their Impact on Students' Depression Level and Academic Performance*. FRONTIERS IN PSYCHIATRY, 2022. **13**.
9. Chen, T.H. and M. Lucock, *The mental health of university students during the COVID-19 pandemic: An online survey in the UK*. PLOS ONE, 2022. **17**(1).
10. Lydiard, R.B., *The role of GABA in anxiety disorders*. JOURNAL OF CLINICAL PSYCHIATRY, 2003. **64**: p. 21–27.
11. Kok, A., *Cognitive control, motivation and fatigue: A cognitive neuroscience perspective*. BRAIN AND COGNITION, 2022. **160**.
12. Almarzouki, A.F., *Stress, working memory, and academic performance: a neuroscience perspective*. STRESS-THE INTERNATIONAL JOURNAL ON THE BIOLOGY OF STRESS, 2024. **27**(1).
13. Xiong, Q., et al., *Factors influencing fatigue, mental workload and burnout among Chinese health care workers during public emergencies: an online cross-sectional study*. BMC NURSING, 2024. **23**(1).
14. Brazaitis, M. and A. Satas, *Regular short-duration breaks do not prevent mental fatigue and decline in cognitive efficiency in healthy young men during an office-like simulated mental working day: An EEG study*. INTERNATIONAL JOURNAL OF PSYCHOPHYSIOLOGY, 2023. **188**: p. 33–46.
15. Jagannathan, R., et al., *Global Updates on Cardiovascular Disease Mortality Trends and Attribution of Traditional Risk Factors*. CURRENT DIABETES REPORTS, 2019. **19**(7).
16. Zhou, B., et al., *Worldwide trends in hypertension prevalence and progress in treatment and control from 1990 to 2019: a pooled analysis of 1201 population-representative studies with 104 million participants*. LANCET, 2021. **398**(10304): p. 957–980.
17. Spruill, T.M., et al., *Association Between High Perceived Stress Over Time and Incident Hypertension in Black Adults: Findings From the Jackson Heart Study*. JOURNAL OF THE AMERICAN HEART ASSOCIATION, 2019. **8**(21).
18. Haight, B.L., et al., *Combined Effects of Cumulative Stress and Daily Stressors on Daily Health*. HEALTH PSYCHOLOGY, 2023. **42**(5): p. 325–334.
19. Alsulaihebi, H.S., et al., *Musculoskeletal disorder prevalence and its correlation with stress in medical students: A cross sectional survey*. JOURNAL OF FAMILY MEDICINE AND PRIMARY CARE, 2024. **13**(4): p. 1524–1529.
20. Gilchrist, A. and A. Pokorná, *Prevalence of musculoskeletal low back pain among registered nurses: Results of an online survey*. JOURNAL OF CLINICAL NURSING, 2021. **30**(11-12): p. 1675–1683.
21. Eijkelhof, B.H.W., et al., *The effects of workplace stressors on muscle activity in the neck-shoulder and forearm muscles during computer work: a systematic review and meta-analysis*. EUROPEAN JOURNAL OF APPLIED PHYSIOLOGY, 2013. **113**(12): p. 2897–2912.
22. Sanzarello, I., et al., *Central sensitization in chronic low back pain: A narrative review*. JOURNAL OF BACK AND MUSCULOSKELETAL REHABILITATION, 2016. **29**(4): p. 625–633.
23. Garcia-Hernandez, A., P. de la Coba, and G.A.R. del Paso, *Central sensitisation to pain and autonomic deficiencies in fibromyalgia*. CLINICAL AND EXPERIMENTAL RHEUMATOLOGY, 2022. **40**(6): p. 1202–1209.
24. Ferrillo, M., et al., *Pain Management and Rehabilitation for Central Sensitization in Temporomandibular Disorders: A Comprehensive Review*. INTERNATIONAL JOURNAL OF MOLECULAR SCIENCES, 2022. **23**(20).
25. Pedrosa-Naudin, M.A., et al., *Non-Adherence to Antidepressant Treatment and Related Factors in a Region of Spain: A Population-Based Registry Study*. Pharmaceutics, 2022. **14**(12).

26. Tan, P., L.H. Lester, and A.M. Lin, *Music therapy treatments in an inpatient setting-A randomized pilot study*. ARTS IN PSYCHOTHERAPY, 2020. **69**.
27. Hammer, C.M., et al., *Positive effects of medical hypnosis on test anxiety in first year medical students*. INNOVATIONS IN EDUCATION AND TEACHING INTERNATIONAL, 2022. **59**(4): p. 472–482.
28. Hürlimann, P., et al., *Cognitive behavioural therapy to treat stress and insomnia: A randomized wait list-controlled trial of two online courses*. JOURNAL OF SLEEP RESEARCH, 2023. **32**(4).
29. Chakales, P.A., J. Locklear, and T. Wharton, *Medicine and Horsemanship: The Effect of Equine-assisted Activities and Therapies on Stress and Depression in Medical Students*. CUREUS JOURNAL OF MEDICAL SCIENCE, 2020. **12**(2).
30. Ju, M.S., et al., *Effects of Aroma Massage on Home Blood Pressure, Ambulatory Blood Pressure, and Sleep Quality in Middle-Aged Women with Hypertension*. EVIDENCE-BASED COMPLEMENTARY AND ALTERNATIVE MEDICINE, 2013. **2013**.
31. Dilaveri, C.A., et al., *Massage Compared with Massage Plus Acupuncture for Breast Cancer Patients Undergoing Reconstructive Surgery*. JOURNAL OF ALTERNATIVE AND COMPLEMENTARY MEDICINE, 2020. **26**(7): p. 602–609.
32. St Martin, M.W., Z. Vidic, and R. Oxhandler, *Exploring the impact of a mindfulness meditation class on college student stress levels and quality of life*. JOURNAL OF AMERICAN COLLEGE HEALTH, 2022.
33. Marshall, M., et al., *A Comparison of the Acute Effects of Different Forms of Yoga on Physiological and Psychological Stress: A Pilot Study*. INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH, 2020. **17**(17).
34. Weerapong, P., P.A. Hume, and G.S. Kolt, *The mechanisms of massage and effects on performance, muscle recovery and injury prevention*. Sports Med, 2005. **35**(3): p. 235–56.
35. Nelson, N.L., *Massage therapy: understanding the mechanisms of action on blood pressure. A scoping review*. J Am Soc Hypertens, 2015. **9**(10): p. 785–793.
36. Brummitt, J., *The role of massage in sports performance and rehabilitation: current evidence and future direction*. N Am J Sports Phys Ther, 2008. **3**(1): p. 7–21.
37. Mori, H., et al., *Effect of massage on blood flow and muscle fatigue following isometric lumbar exercise*. Medical science monitor : international medical journal of experimental and clinical research, 2004. **10**: p. CR173–8.
38. Zhang, L., et al., *Evaluating the role of Shujing Tongdu massage in enhancing wound recovery and alleviating spinal disease symptoms: A randomized controlled trial*. INTERNATIONAL WOUND JOURNAL, 2024. **21**(1).
39. Field, T., et al., *Cortisol decreases and serotonin and dopamine increase following massage therapy*. INTERNATIONAL JOURNAL OF NEUROSCIENCE, 2005. **115**(10): p. 1397–1413.
40. Hernandez-Reif, M., et al., *Migraine headaches are reduced by massage therapy*. INTERNATIONAL JOURNAL OF NEUROSCIENCE, 1998. **96**(1-2): p. 1–11.
41. Melzack, R. and P.D. Wall, *Pain mechanisms: a new theory*. Science, 1965. **150**(3699): p. 971–9.
42. Kim, S.K., et al., *Clinical outcomes and cost-effectiveness of massage chair therapy versus basic physiotherapy in lower back pain patients A randomized controlled trial*. MEDICINE, 2020. **99**(12).
43. Cohen, S., T. Kamarck, and R. Mermelstein, *A GLOBAL MEASURE OF PERCEIVED STRESS*. JOURNAL OF HEALTH AND SOCIAL BEHAVIOR, 1983. **24**(4): p. 385–396.
44. Li, T.J., et al., *Design, Analysis, and Reporting of Crossover Trials for Inclusion in a Meta-Analysis*. PLOS ONE, 2015. **10**(8).
45. Podlesek, A., L. Komidar, and V. Kavcic, *The Relationship Between Perceived Stress and Subjective Cognitive Decline During the COVID-19 Epidemic*. FRONTIERS IN PSYCHOLOGY, 2021. **12**.
46. Li, G., et al., *The impact of mental fatigue on brain activity: a comparative study both in resting state and task state using EEG*. BMC NEUROSCIENCE, 2020. **21**(1).
47. Zhang, H., et al., *Objective Assessments of Mental Fatigue During a Continuous Long-Term Stress Condition*. FRONTIERS IN HUMAN NEUROSCIENCE, 2021. **15**.
48. Jap, B.T., et al., *Using EEG spectral components to assess algorithms for detecting fatigue*. EXPERT SYSTEMS WITH APPLICATIONS, 2009. **36**(2): p. 2352–2359.
49. Lim, J.-H., et al., *The effects on mental fatigue and the cognitive function of mechanical massage and binaural beats (brain massage) provided by massage chairs*. Complementary Therapies in Clinical Practice, 2018. **32**: p. 32–38.
50. Moraska, A., et al., *Physiological Adjustments to Stress Measures Following Massage Therapy: A Review of the Literature*. EVIDENCE-BASED COMPLEMENTARY AND ALTERNATIVE MEDICINE, 2010. **7**(4): p. 409–418.
51. Morhenn, V., L.E. Beavin, and P.J. Zak, *Massage Increases Oxytocin and Reduces Adrenocorticotropin Hormone in Humans*. ALTERNATIVE THERAPIES IN HEALTH AND MEDICINE, 2012. **18**(6): p. 11–18.

52. Light, K.C., K.M. Grewen, and J.A. Amico, *More frequent partner hugs and higher oxytocin levels are linked to lower blood pressure and heart rate in premenopausal women*. BIOLOGICAL PSYCHOLOGY, 2005. **69**(1): p. 5–21.
53. Park, Y.K., et al., *Effects of Self-acupunctural Hand Massage Using Aromatic Oil on Alleviating the Stress, Fatigue, and Vital Signs of Korean Middle-Aged Women*. HOLISTIC NURSING PRACTICE, 2022. **36**(4): p. 247–254.
54. Akbas, E., B. Ünver, and E.U. Erdem, *Acute Effects of Connective Tissue Manipulation on Autonomic Function in Healthy Young Women*. COMPLEMENTARY MEDICINE RESEARCH, 2019. **26**(4): p. 250–257.
55. Lee, Y.M. and Y.R. Yeun, *Effects of Combined Foot Massage and Cognitive Behavioral Therapy on the Stress Response in Middle-Aged Women*. The Journal of Alternative and Complementary Medicine, 2017. **23**(6): p. 445–450.
56. Billhult, A., et al., *The effect of massage on immune function and stress in women with breast cancer - A randomized controlled trial*. AUTONOMIC NEUROSCIENCE-BASIC & CLINICAL, 2009. **150**(1-2): p. 111–115.
57. Lundstam, U., et al., *Epidermal growth factor levels are related to diastolic blood pressure and carotid artery stiffness*. SCANDINAVIAN CARDIOVASCULAR JOURNAL, 2007. **41**(5): p. 308–312.
58. Olney, C.M., *The effect of therapeutic back massage in hypertensive persons: A preliminary study*. BIOLOGICAL RESEARCH FOR NURSING, 2005. **7**(2): p. 98–105.
59. Cho, H.A., *Effects of Self-Efficacy Information and Self-Hand Massage on Psychosocial Factors and Smoking Cessation in South Korean Adolescents*. JOURNAL OF CHILD & ADOLESCENT SUBSTANCE ABUSE, 2018. **27**(1): p. 29–41.
60. Luck, J.C., et al., *Multiple Laser Doppler Flowmetry Probes Increase the Reproducibility of Skin Blood Flow Measurements*. FRONTIERS IN PHYSIOLOGY, 2022. **13**.
61. Moghadas, M., et al., *Cardiorespiratory effect of Swedish back massage in hypertensive patients: a randomized clinical trial*. ARTERIAL HYPERTENSION, 2021. **25**(4): p. 159–163.
62. Loh, E., et al., *Effect of progressive muscle relaxation on postoperative pain, fatigue, and vital signs in patients with head and neck cancers: A randomized controlled trial*. PATIENT EDUCATION AND COUNSELING, 2022. **105**(7): p. 2151–2157.
63. Bryan, N.S., *Nitric oxide deficiency is a primary driver of hypertension*. BIOCHEMICAL PHARMACOLOGY, 2022. **206**.
64. Kaada, B. and O. Torsteinbo, *INCREASE OF PLASMA BETA-ENDORPHINS IN CONNECTIVE-TISSUE MASSAGE*. GENERAL PHARMACOLOGY-THE VASCULAR SYSTEM, 1989. **20**(4): p. 487–489.
65. Goldberg, J., S.J. Sullivan, and D.E. Seaborne, *THE EFFECT OF 2 INTENSITIES OF MASSAGE ON H-REFLEX AMPLITUDE*. PHYSICAL THERAPY, 1992. **72**(6): p. 449–457.
66. Palmieri, R.M., C.D. Ingersoll, and M.A. Hoffman, *The Hoffmann reflex: Methodologic considerations and applications for use in sports medicine and athletic training research*. JOURNAL OF ATHLETIC TRAINING, 2004. **39**(3): p. 268–277.
67. Young, J.D., A.J. Spence, and D.G. Behm, *Roller massage decreases spinal excitability to the soleus*. JOURNAL OF APPLIED PHYSIOLOGY, 2018. **124**(4): p. 950–959.
68. Chen, P.C., et al., *The Effect of Massage Force on Relieving Nonspecific Low Back Pain: A Randomized Controlled Trial*. INTERNATIONAL JOURNAL OF ENVIRONMENTAL RESEARCH AND PUBLIC HEALTH, 2022. **19**(20).
69. Myers, M.G., M. Valdivieso, and A. Kiss, *Use of automated office blood pressure measurement to reduce the white coat response*. J Hypertens, 2009. **27**(2): p. 280–6.

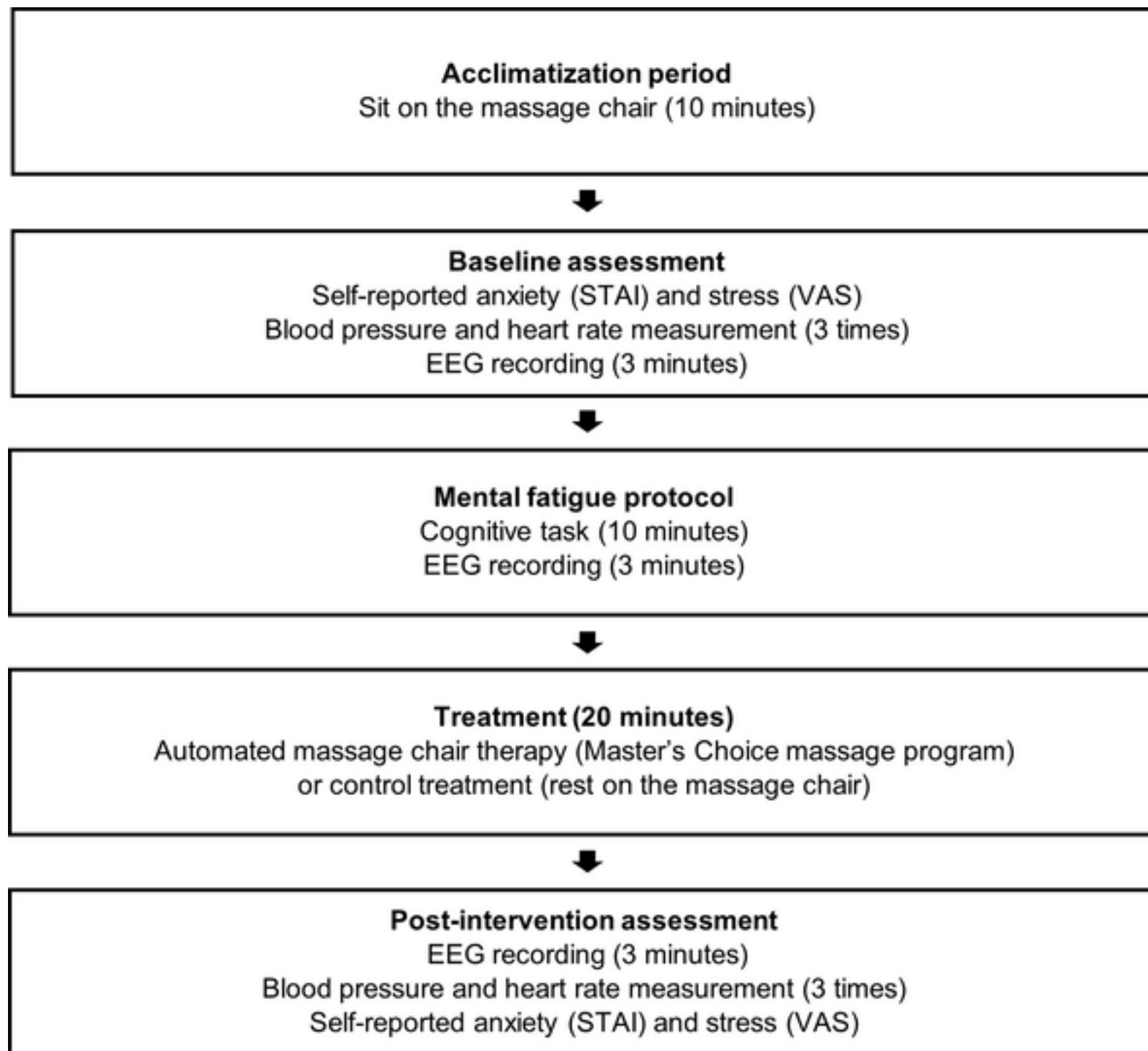


Figure 2.2: Participant resting on the massage chair while EEG is recorded

[Click here to access/download;Figure \(300 dpi and editable format\);Figure 2.2.jpg](#)



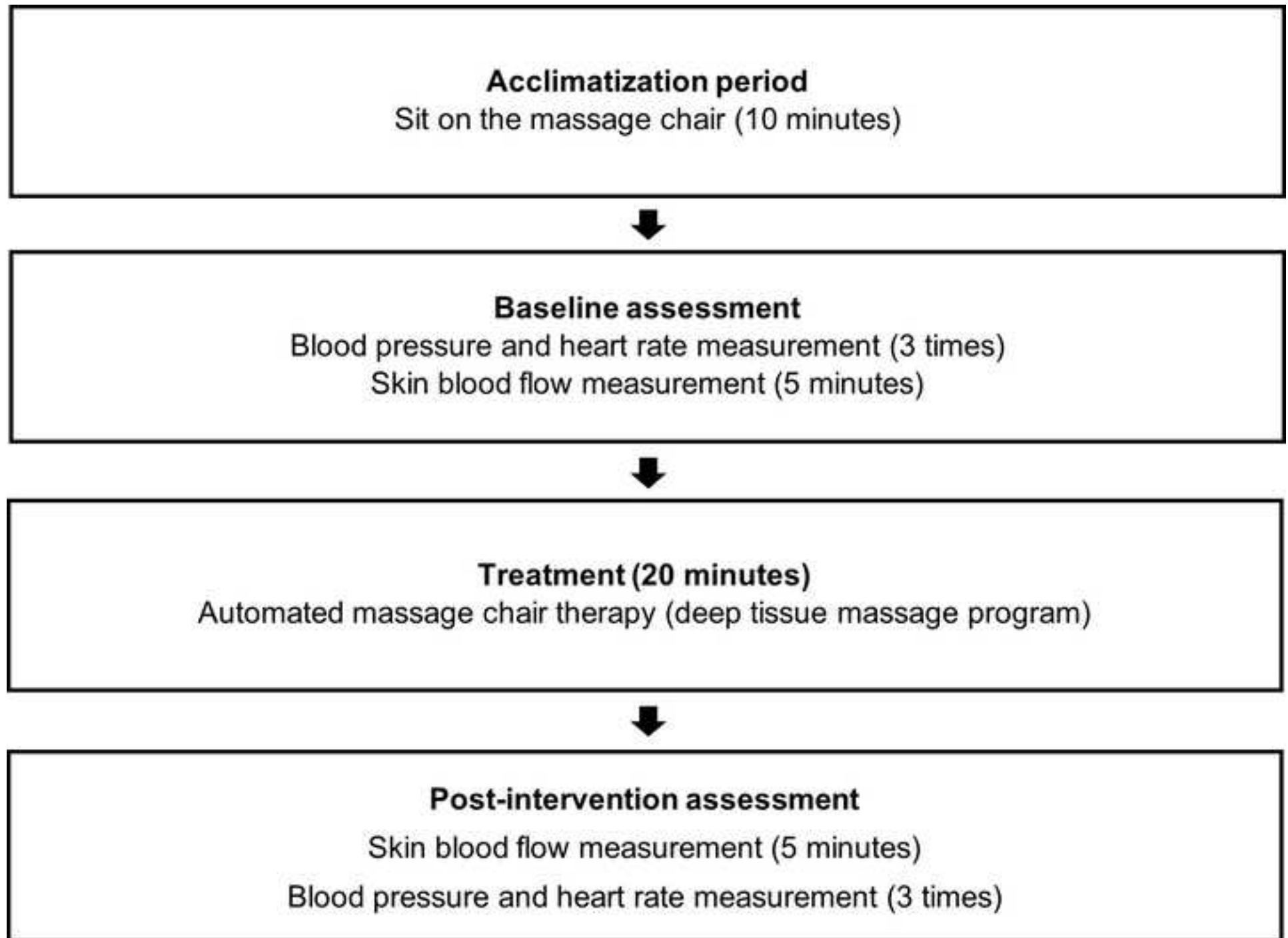


Figure 3.2: a) form factor of the probe b) placement of the probe

[Click here to access/download;Figure \(300 dpi and editable format\);Figure 3.2.jpg](#) 



Figure 3.3: Bland-Altman plot of skin blood flow for session 1 between pre and post measurement (n = 19)

[Click here to access/download;Figure \(300 dpi and editable format\);Figure 3.3.jpg](#)

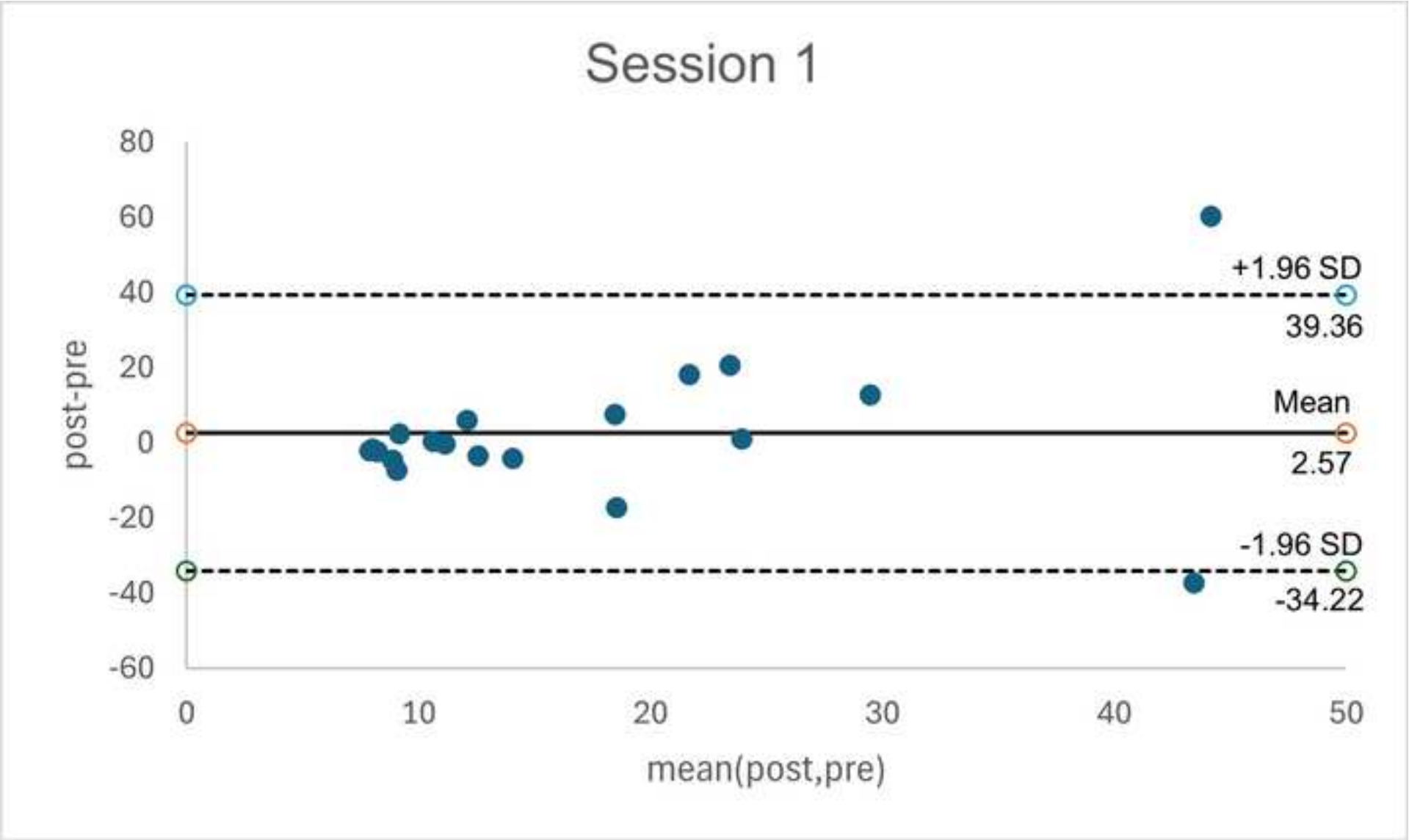


Figure 3.4: Bland-Altman plot of skin blood flow for session 2 between pre and post measurement (n = 19)

[Click here to access/download;Figure \(300 dpi and editable format\);Figure 3.4.jpg](#)

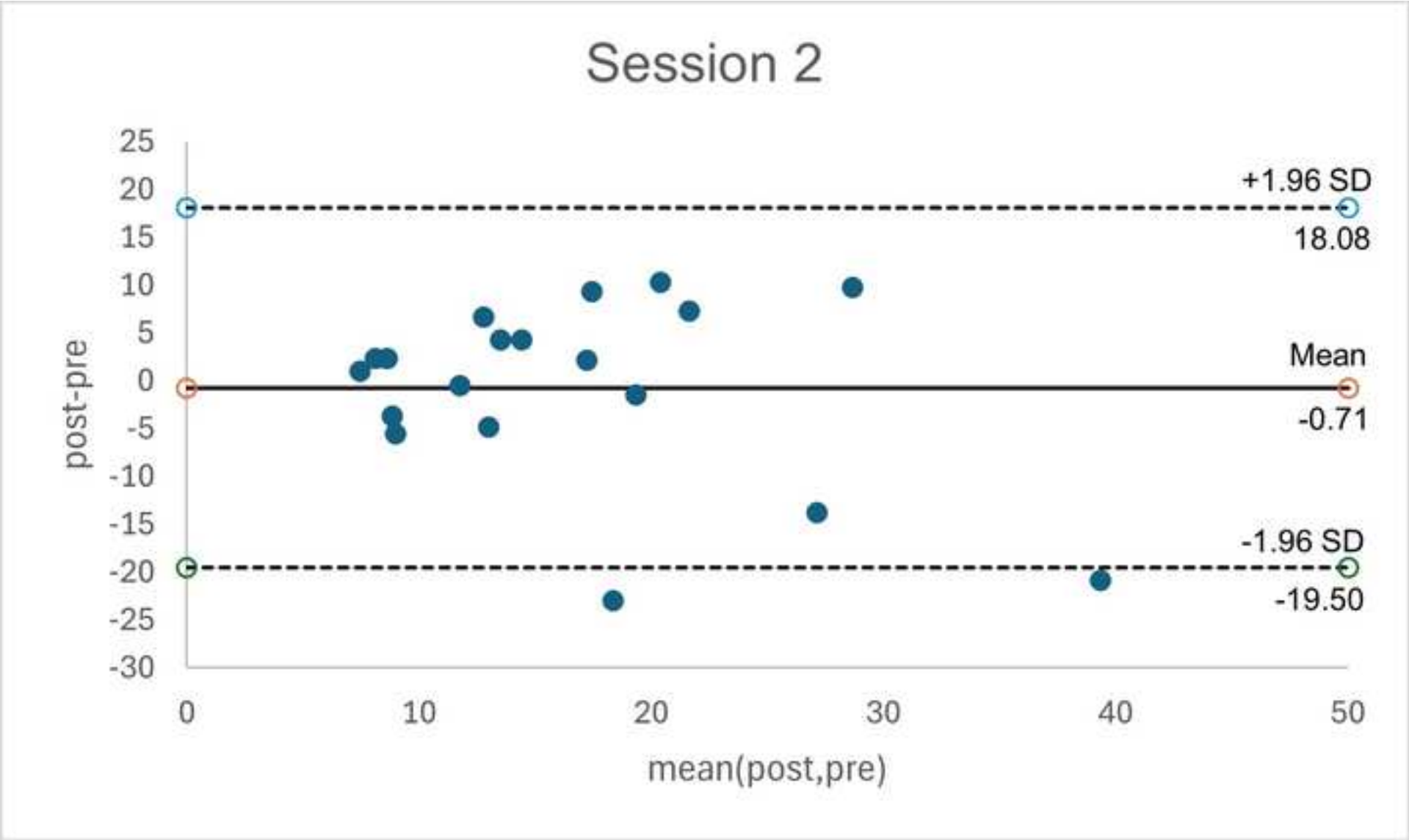
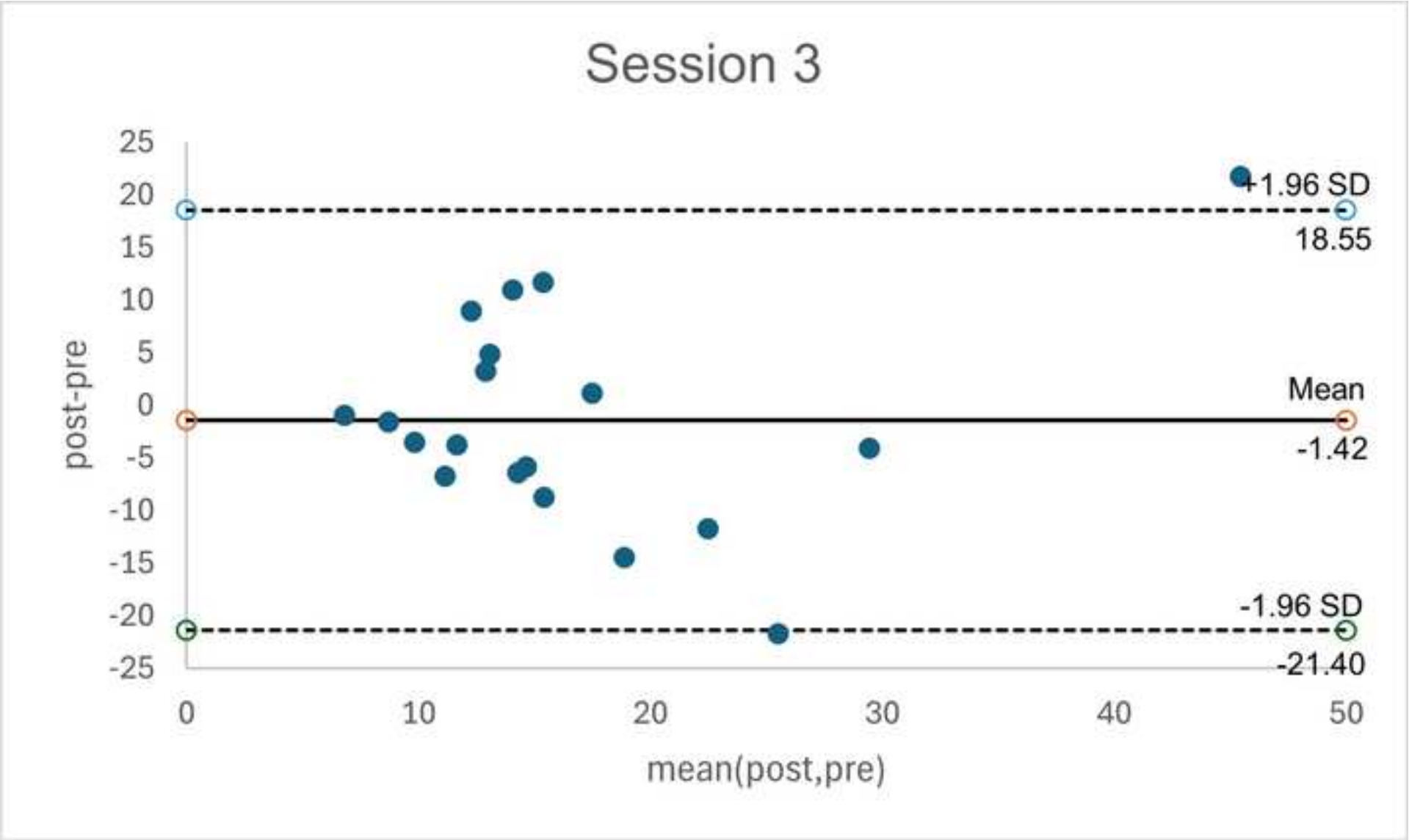
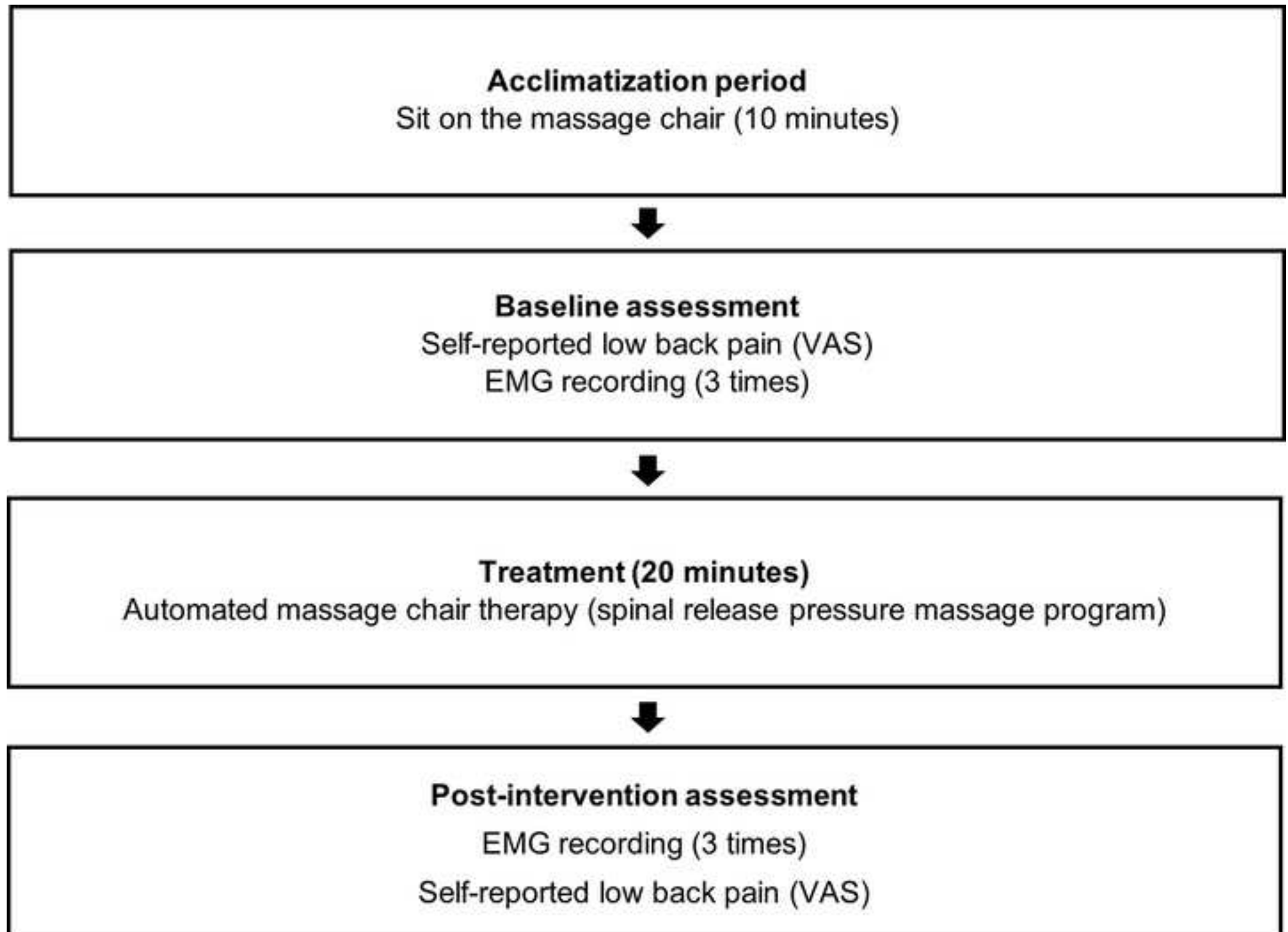


Figure 3.5: Bland-Altman plot of skin blood flow for session 3 between pre and post measurement (n = 19)

[Click here to access/download;Figure \(300 dpi and editable format\);Figure 3.5.jpg](#)





Declaration of Competing Interest

The author(s) declare that they have no conflicts of interest.

CRediT authorship contribution statement

Xin Lee: Conceptualization, Methodology, Visualization, Formal analysis, Investigation, Writing – Original Draft, Data Curation. **Juliana Usman:** Resources, Supervision. **Norhamizan Hamzah:** Methodology, Resources, Supervision, Validation. **Pauline Shan Qing Yeoh:** Methodology, Validation, Writing – Review and Editing. **Xiang Wu:** Funding acquisition, Supervision. **Joon Huang Chuah:** Supervision. **Khin Wee Lai:** Supervision, Writing – Review and Editing, Validation, Project administration, Funding acquisition.

Author Agreement Declaration

We, the undersigned authors, hereby declare that:

1. Authorship and Contribution

- All authors listed have made substantial contributions to the conception, design, data acquisition, analysis, and interpretation of the study.
- Each author has participated sufficiently in the work to take public responsibility for appropriate portions of the content.
- All authors have reviewed and approved the final manuscript.

2. Originality and Prior Publication

- The manuscript is original and has not been published elsewhere.
- The manuscript is not under consideration for publication by any other journal.

3. Conflict of Interest

- All authors have disclosed any potential conflicts of interest, including financial, personal, or professional relationships that could influence the work.
- If no conflicts exist, the statement "None declared" is included.

4. Ethical Approval

- For studies involving human participants, appropriate ethical approval has been obtained from a recognized ethics committee or institutional review board.
- For studies involving animals, ethical approval has been obtained in accordance with institutional and national guidelines.

5. Informed Consent

- Informed consent has been obtained from all participants involved in the study.

6. Data Availability

- The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

7. Copyright and Licensing

- Upon acceptance, the authors agree to transfer copyright to the publisher or grant a license for publication, as per the journal's policies.

8. Corresponding Author

- The corresponding author is responsible for all communications with the journal, including manuscript submission, revisions, and final approval.

Signatures:


Dr. Nizamizwan Hamzah
Department of Neurosurgery
Faculty of Medicine
University of Malaya
50603 Kuala Lumpur


Dr. Juliana Usman
Senior Lecturer
Department of Biomedical Engineering
Faculty of Engineering
University of Malaya
50603 Kuala Lumpur, Malaysia


Assoc. Prof. Ir. Dr. Chuah Joon Huang
Associate Professor
Department of Electrical Engineering
Faculty of Engineering
University of Malaya
50603 Kuala Lumpur.


Prof. Madya. Hs. Dr. Lai Khin Wee
Jabatan Kejuruteraan Bioperubatan
Fakulti Kejuruteraan
Universiti Malaya
Emel : lai.khinwee@um.edu.my
Tel : 03-79677627